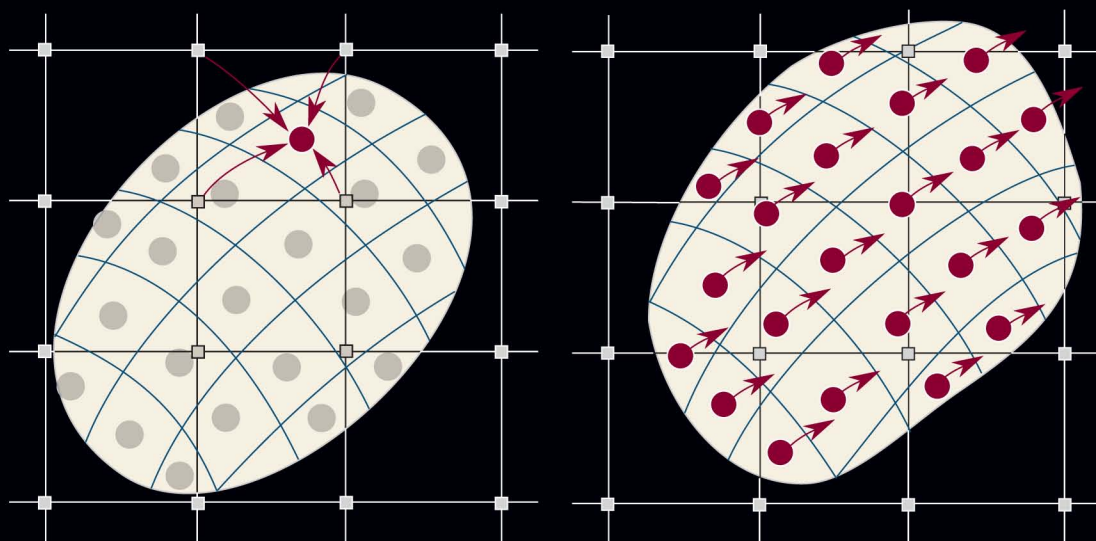


The Material Point Method for Geotechnical Engineering

A Practical Guide

EDITED BY

James Fern • Alexander Rohe
Kenichi Soga • Eduardo Alonso



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Foreword

Geotechnical engineering differs from other forms of engineering by the composition of ground, which is composed of a wide range of solids with discontinuities, pores fluids and structures. These different elements composing the ground interact with one another increasing the complexity of any prediction. In traditional geotechnical analysis, limit equilibrium methods (LEM) are useful techniques for estimating the onset of failure. However, these simple methods cannot provide any information on the deformation and are unable to simulate the real soil behaviour. Other well-established methods, such as the Finite Element Method (FEM), can model complex geometries and soil behaviours. However, FEM is limited to small deformation analyses and cannot model post-failure behaviours. New numerical methods are being developed in order to provide an engineering tool capable of simulating the entire instability process from the static stability analysis at small deformation to the dynamic post-failure run-out behaviour. This is the case of the material point method (MPM). MPM has been applied to a number of geotechnical problems and it has been extended to solve coupled flow-deformation problems in saturated conditions as it can simultaneously model fluid-like and solid-like behaviours. Fig. 1 depicts different applications of MPM in geotechnical engineering.

Sound geotechnical engineering practice is about predicting the most reliable behaviours of infrastructure during construction, operational use, maintenance, and deconstruction with adequate risk management. As pointed out by Dr Suzanne Lacasse in her Rankine Lecture, adding a probabilistic analysis and risk assessment enables assessing the inherent uncertainties in soil and material geometries and properties, probable loading conditions and numerical assumptions. Geotechnical engineering practices of simulation-based design then apply sufficiently large safety factors to bring risks to an acceptable level. This results in reliable and up-to-standard design of buildings, constructions and infrastructures with a very low probability of failure. However, excessive deformation, and even failure, of physical assets due to extreme events rarely occur. The often-unanticipated extreme events may be caused by exceptional ground or water conditions, extreme loading situations outside the design values, and external impacts of falling or penetrating objects as well as human errors during design and construction. Asset owners and managers increasingly demand quantifying in advance these improbable large deformation or failure events and associated risks in order to design appropriate counter measures to prevent them or to reduce their effects. Dr Evert Hoek pointed out in his 2014 lecture series: "There is no point in calculating the probability of failure if we

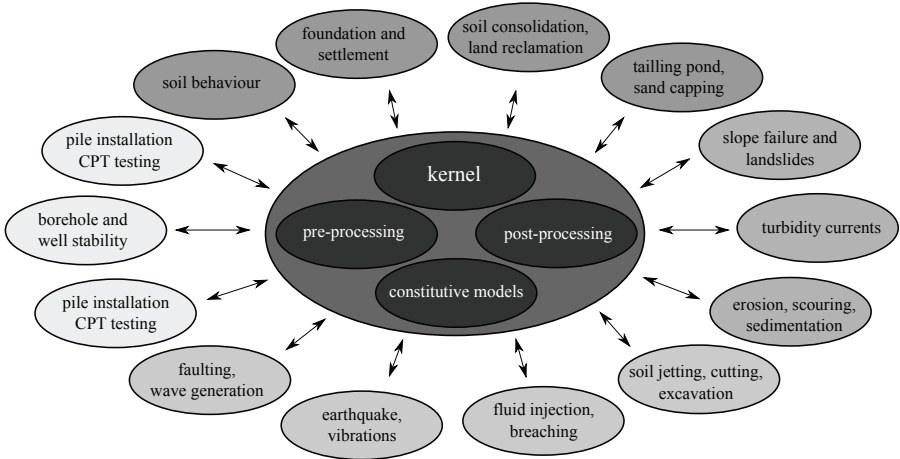


FIGURE 1: Range of applications of MPM in geotechnical engineering.

do not know what the consequence of that failure will be". MPM provides means of investigating the low-probability but high risk extreme events.

Making tools available will allow extending current geotechnical risk analysis to practice. It will firstly result in better contingency planning and measures and it will help to inform stakeholders and citizens on the geotechnical risks, potential damage and available mitigation measures. Secondly, business opportunities are currently missed to design more efficient infrastructure, decrease safety factors and reduce costs. For various structures, large deformations during their life cycle are acceptable when safety is guaranteed, damage is not disrupting construction and repair is affordable. These business opportunities become reality when a proper modelling approach and tools are available.

Ipo L. Ritsema and Simone van Schijndel
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--- *List of Common Symbols* ---

Symbol Description

Acronyms

BSMPM	B-spline material point method
BC	boundary condition
CC	Cam-Clay
CG	conjugate gradient
CPDI	convected particle domain interpolation
CPT	cone penetration test
CPTu	piezocone penetration test
CS	critical state
CSL	critical state line
CSSM	critical state soil mechanics
DDMP	dual domain material point method
DOF	degree of freedom
FEM	finite element method
GIMP	generalised interpolation material point
HP	hypoplastic model
IBM	implicit boundary method
IgA	isogeometric analysis
LAM	limit analysis method
LAS	local average subdivision
LEM	limit equilibrium method
LMP	liquid material point
MP	material point
MPM	material point method
RMPM	random material point method
MC	Mohr-Coulomb
MCSS	Mohr-Coulomb strain softening
MCC	modified Cam-Clay

NCL	virgin compression line
NMD	nodal mixed discretisation
MUSL	modified updated stress last
NS	Nor-Sand
OCC	original Cam-Clay
OCR	overconsolidation ratio
PIC	particle in-cell method
SQPSO	species-based quantum particle swarm optimisation
RBD	reliability based design
RMS	root mean square
SWE	shallow water equation
SWRC	soil water retention curve
SPD	symmetric positive definite
SMP	solid material point
THM	thermo-hydro-mechanical
TLS	Taylor least square
ULFEM	updated Lagrangian finite element method
VOF	volume of fluid

Roman symbols

a	aspect ratio
$\vec{a}$	acceleration
A^e	shear modulus constant
A_{ref}	reference area
$\vec{b}$	body force
c	cohesion
c_h	horizontal consolidation coefficient
c_L	specific heat of liquid
c_{NB}	Courant number

c_s	specific heat of solid	K_0	earth pressure coefficient at rest
c_u	undrained strength	m	mass
c_v	vertical consolidation coefficient	$\vec{m}$	diagonal of lumped mass matrix
D	dilatancy rate	M	critical state stress ratio
D_{crit}	critical depth for excavations	$\mathbf{M}$	(lumped) mass matrix
$\mathbf{D}$	stiffness matrix	$\mathbf{M}^C$	consistent mass matrix
e	void ratio	n	porosity
E	Young modulus	n^e	shear modulus exponent
E_u	undrained Young modulus	N	model parameter, dilatancy coefficient, or NCL intercept
E_a^p	accumulated plastic strain	N_g	total number of Gauss points
$\vec{f}$	force	N_n	total number of DOFs
$\tilde{f}$	least-square approximation of function	p'	Cambridge mean effective stress
$\bar{f}$	volume average of function	$\vec{p}$	momentum vector
$f_1^{\text{B-M}}$	liquid flow rate between band and matrix	p_c	preconsolidation pressure
$f_\theta^{\text{B-M}}$	heat flow rate between band and matrix	p_i	image pressure
f_s	CPT sleeve friction	p_g	pore gas pressure
F	yield function	p_l	pore liquid pressure
g	gravity	p_w	pore water pressure
G	shear modulus	δp_w	excess pore water pressure
G_s	specific gravity	P	potential function
h	height	q	Cambridge deviatoric stress
h^*	normalised height	q_c	CPT cone resistance
H	hardening modulus	q_h	heat flow conduction
$\dot{H}$	heat rate	$\vec{q}_l$	Darcy flow vector
i	hydraulic gradient	Q	crushing pressure
I_C	relative pressure index	$\vec{r}$	residual vector
I_D	relative density index	r^*	normalised run-out distance
I_L	liquidity index	R	fitting parameter
I_L^{eq}	equivalent liquidity index	s	matrix suction
I_p	plasticity index	S_r	degree of saturation
I_r	rigidity index	t	time
I_R	relative dilatancy index	T	temperature
$\mathbf{I}$	identity matrix	T^*	modified normalised time
$\vec{I}$	identity vector	u	displacement
J	invariant of deviatoric stress tensor	u_1	pore water pressure at cone face
$\mathbf{J}$	Jacobian matrix		
k	hydraulic conductivity		
K	bulk modulus		

u_2	pore water pressure at cone shoulder	θ	Lode angle
u_3	pore water pressure at cone friction sleeve	θ_h	horizontal spatial variability
UCS	uniaxial compressive strength	θ_v	vertical spatial variability
v	specific volume	Θ	temperature
$\vec{v}$	velocity vector	κ	swelling modulus
v_0	maximal initial velocity	λ	CSL slope
v_{mp}	material point velocity	λ_{max}	maximum eigenvalue
V	volume	λ_{min}	minimum eigenvalue
V	normalised penetration velocity	Λ	plastic multiplier
w	water content	μ	friction coefficient
W	work	μ_1	liquid viscosity
$\dot{W}$	work rate	μ_w	water viscosity
W_p	history variable	ν	Poisson ratio
x	global coordinate	ρ	unit mass
y	global coordinate	σ	stress
z	global coordinate	σ^r	radial stress
		σ^{tot}	total stress
		σ'	effective stress
		τ	shear stress
		τ_f	shear strength
		φ	friction angle
		ϕ	basis function vector
		χ	dilatancy coefficient
		ϕ	basis function vector
		ψ	dilatancy angle
		ψ_i	Taylor basis function
		ψ_l	liquid transfer coefficient
		ψ_θ	heat transfer coefficient
		Ψ	state parameter
		Ξ	knot vector
		ξ_i	knot
		ω_g	Gauss weight
		Ω	domain
<i>Greek symbols</i>			
α	damping factor, or dilatancy coefficient		
α_1	liquid compressibility		
β	volumetric thermal expansion coefficient, or mass scaling factor		
γ	unit weight, shear strain		
$\dot{\gamma}$	shear strain rate		
Γ	CSL intercept, or thermal conductivity coefficient		
δ	increment		
$\dot{\delta}$	displacement rate		
$\delta_{i,j}$	Dirac delta function		
ε	strain		
ε_{dev}	deviatoric strain		
ε_r	radial strain		
ε_{vol}	volumetric strain		
ε_θ	circumferential strain		
ζ	softening shape factor		
η'	effective stress ratio		
η_w	water surface elevation		
<i>Superscripts and subscripts</i>			
		B	shear band
		c	consolidation
		cs	critical state
		dev	deviatoric
		e	elastic
		ext	external
		eq	equivalent
		f	failure

g or G	gas	sat	saturated
i	image	t	time
int	internal	tc	triaxial compression
l or L	liquid	te	triaxial extension
m	mixture	T	transpose
mp	material point	u	undrained
max	maximum	v or V	void
min	minimum	vol	volumetric
M	matrix	w	water
N	normal	x	x-direction
p	plastic	y	y-direction
q	deviatoric stress	z	z-direction
res	residual	'	effective
s or S	solid		

Part I

Theory



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E.J. Fern, K. Soga and E.E. Alonso

1.1 Introduction

The art of computational geomechanics refers to the process of computing the most probable outcomes of a given engineering problem. The science of computational geomechanics refers to the computation itself and it ensures that the predictions are exempt of computational errors. Whilst the science of computational geomechanics is well documented in the literature (i.e. [231, 232, 348]), the art of computational geomechanics is not so well documented; the document by Potts et al. [229] is one of the most comprehensive guideline documents available.

In order to carry out a prediction, a series of choices and idealisation have to be made and these choices are based on engineering judgement. Computational tools are powerful engineering aids, which open up new engineering opportunities, but they are not replacements for blind judgement. Although there has been great progress in computational geomechanics in the past years, some major issues still remain. These include data and knowledge deficiencies and limited computer capacity [134]. Computational geomechanics differs from other forms of computational mechanics by the complex behaviour of geomaterials and its spatial variability. A thorough understanding of geomechanics is necessary to carry out any numerical predictions in geotechnical engineering [263].

Potts [230] raised the following question in his Rankine Lecture: "Is numerical analysis a virtual dream or a practical reality?" This question was motivated by the lack of popularity of numerical predictions in practice at that time. The lack of knowledge, know-how and guidance in computational geomechanics can hamper the use of numerical methods in practice. Although many engineers have been exposed to numerical methods during their education or career, the complexity of an analysis and its practical potential are not always fully appreciated [230].

The increasing computational capacity of personal and high-performance computers allows the development of new numerical methods, which permits modelling not only small deformation problems but also large deformation ones. Alonso demonstrated in his Rankine Lecture that these new methods could be used to explain the triggering mechanism and motion of landslides. This book aims to provide guidance on the use of the material point method (MPM) for geotechnical engineering. This chapter outlines the process of carrying out numerical predictions. Although this book is dedicated to MPM, some elements presented in this chapter are applicable to other numerical methods.

1.2 General context

Numerical analyses are time consuming and expensive but they permit exploring problems with complicated geometries, complex material behaviours, variable loading sequences, and interactions between existing and new infrastructure for which conventional methods are incapable of solving. They permit carrying out predictions of a probable outcome considering the interaction between different mechanics and structures (i.e. hydro-mechanical coupling, thermo-hydro-mechanical coupling, soil-structure interactions). Numerical analyses are used (1) to gain additional insight in a given problem and (2) to explore different probable outcomes.

Numerical predictions in geotechnical engineering are important due to the complex mechanical behaviour of geomaterials but they are hampered by the difficulty in making accurate predictions due to inherent variability and uncertainty of the ground conditions [161]. The inaccuracy is not related to the science of computational geomechanics but to the uncertainty of the input parameters. The absence of information on the site conditions is a common and major issue for any prediction. This is due to difficulties in subsurface exploration and its associated cost. However, it does not make computational geomechanics worthless but it points out some deficiency in obtaining the required information.

In order to avoid confusion with the different terms used in this book, the following definitions are adopted.

A model is the geometrical and mechanical idealisation of a given problem. It includes, although not exclusively, the idealised geometry and stratigraphy, the boundary conditions, the contact conditions and the mesh. The model should be simple enough to permit computational-efficient simulations but it must also contain sufficient details to capture the correct mechanics.

A simulation is the actual computation of a specific probable outcome using a given model, a type of analysis, specific constitutive models and various contact conditions. Therefore, different simulations can be generated from

a given model. It includes the computation choices such as defining the critical time step (Courant number), damping conditions and the duration of the simulation. These numerical features are explained in [Chapter 6](#).

An analysis is the whole process of mathematical idealisation of a real problem, including the computation of a prediction and its interpretation.

1.3 Anatomy and classification of predictions

In an attempt to provide guidance, Lambe [161] described the process of carrying out numerical simulations and called it *the anatomy of prediction* (Fig. 1.1). Each step of the process is a seldom discrete isolated component but an interrelationship between steps does exist as the choices made in one step impact the following ones. Therefore, the numerical prediction obtained at the end of the process is dependent on the choices and assumptions made in every step, and on the state of knowledge prior to the computational idealisation. Numerical simulations are not best guesses but approximated solutions of a predefined set of governing equations solved in a specific sequence. The anatomy of prediction serves as the layout of this chapter and each step is discussed in a separate section.

Any numerical prediction relies on the field situation at the time of the prediction. For this reason, Lambe [161] introduced a classification system called *the classification of prediction* (Table 1.1). It is based on the moment the prediction is made and the amount of information available at the time of the prediction.

Class-A predictions are simulation carried out prior to the event and in absence of any field data. This is typically the case for feasibility studies.

Class-B predictions are carried out during the event in absence of any site data. If some site data is available, then the predictions become a class-B1 prediction. This is typically the case for tender designs.

Class-C predictions are carried out after the event of interest but in absence of site data and a class-C1 prediction in presence of site data. These are the most abundant type of simulations in the academic literature as they permit the validation of a given method and assumptions.

Potts and Zdravković [348] give an example of a class-A prediction and a class-C prediction for the stabilisation of the Pisa Tower. The class-A predictions aimed to estimate the number of counter-weights required for the stabilisation and to check undesirable and unexpected responses of the tower. The class-C prediction permitted a better understanding of the ground response.

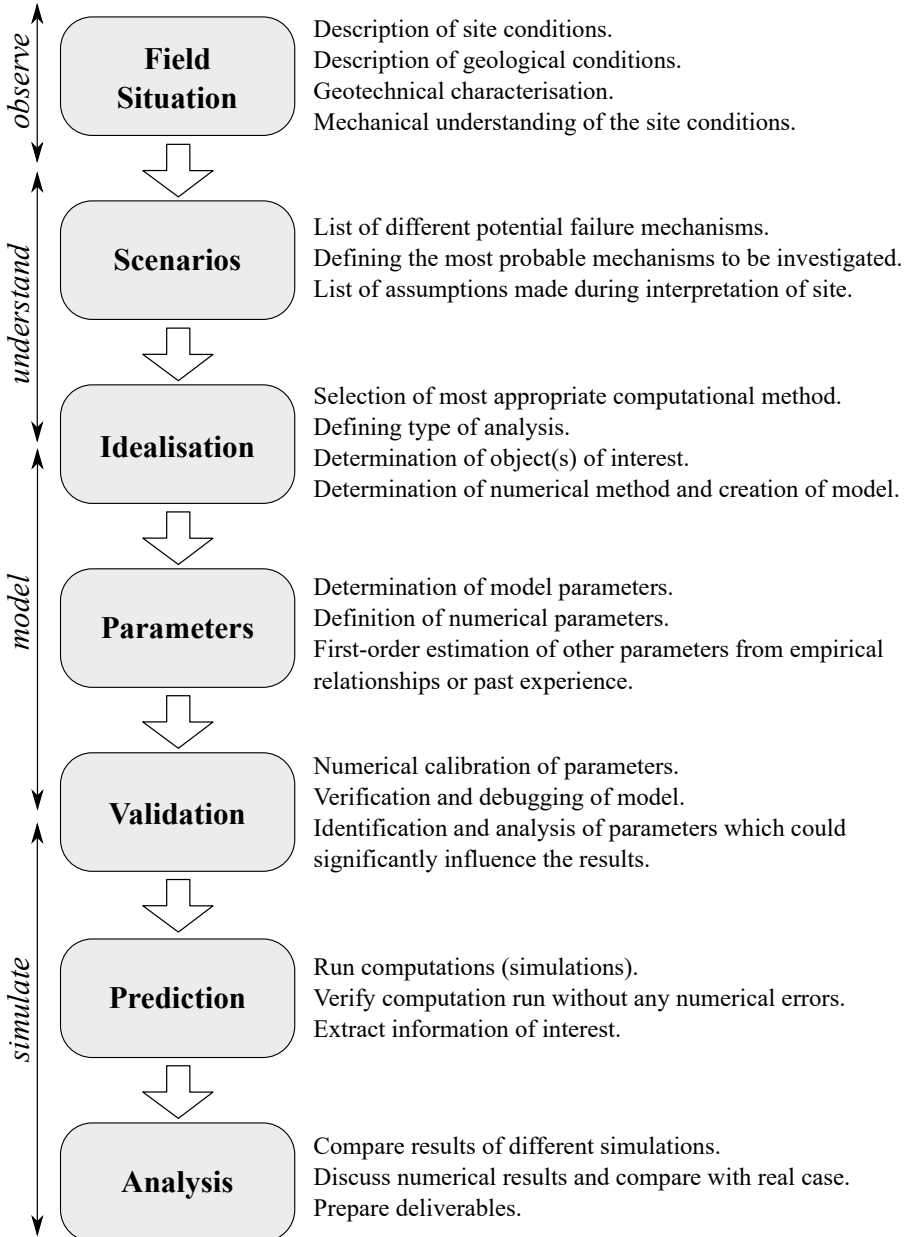


FIGURE 1.1: Anatomy of prediction.

TABLE 1.1
Classification of prediction.

Prediction type	Prediction made	Field results
Class A	before event	-
Class B	during event	unknown
Class B1	during event	known
Class C	after event	unknown
Class C1	after event	known

1.4 Field situation

The first step in any numerical analysis is to describe the field situation, which is composed of the object of interest and its surrounding environment. It may seem at first a trivial step but it is often complex and evolves with time. This is especially the case for design work, which goes through different design stages, and for landslides for which the site investigations can spread over a few years. The content and extent of the field situation is case specific. It can be short and straightforward for simple cases such as a reduced-scale laboratory experiment, for which examples are given in [Chapters 9, 10 and 21](#), but it can also be complex as is the case for the Vajont Landslide ([Chapter 14](#)). The field situation typically includes the following aspects, albeit not exclusively.

The geometry of the object of interest and its surrounding environment have to be described. It is necessary to decide which details of the geometry are relevant for the analysis and which can be neglected. For instance, the description of a tunnel should include the inner diameter, the excavated diameter and the dimensions and types of the linings. It will then be decided which diameter is the most relevant for the analysis, how to model the different linings and if contact conditions or hinges are necessary to model the mechanical behaviour of the lining.

The geological and environmental conditions are used to define the stratigraphy and geotechnical model. The geological model is an idealisation of the location of the different soil and rock units with respect to their history (deposition epoch, constituents, etc.). The geology provides hints on the state of the ground. The geotechnical model is an idealisation of the location of the different soil and rock units in terms of mechanical characteristics. For instance, a given clay can have different stiffness moduli and undrained strengths at different depth. This gives one geological unit but two geotechnical units.

The environmental conditions also need to be described. For example, the presence of hydrocarbons in the ground can decrease the strength of the

soil or rock [174]. Environmental conditions also include variations of the groundwater table associated with seasonal rainfall and groundwater flows.

The geotechnical characterisation describes soil and rock units in terms of geotechnical characteristics. These can be grain-size distribution, texture, Atterberg limits and/or minimum and maximum void ratios. It also describes the mechanical behaviour of different soil and rock units obtained from site and laboratory tests. Based on the geotechnical characterisation, constitutive models are selected and calibrated. Bishop [35] pointed out that errors can be introduced during soil sampling and testing, which do not reproduce sufficiently accurately the soil conditions and stress state of the natural ground. Unless equal attention is paid to each factor, an elaborate mathematical treatment can lead to a fictitious impression of accuracy [35].

The geotechnical characterisation also needs to establish the in-situ stress state, pore pressure profiles, and consolidation states. This includes determining the orientation and magnitude of the principal stresses. This is especially important in tectonically active areas where both horizontal stresses can be larger than the vertical stress giving earth pressure coefficients greater than 1.0 ($\sigma'_{h1} > \sigma'_{h2} > \sigma'_v \rightarrow K_{0,1} > K_{0,2} > 1.0$).

The natural variability of the geotechnical characteristics has encouraged some researchers to include this variability with random fields and carry out statistical analysis for probable outcomes. An example of such an analysis is presented in Chapter 16.

The loading sequence of the object of interest has to be known and clearly described in order to idealise the different stages and loading schemes of a problem. The plastic behaviour of soil and rock is stress-path dependent.

The accuracy of a prediction depends on the quality of the computational method as well as on the data used to make the prediction [161]. The input data is often the weakness of many numerical analyses [134]. Therefore, the quality of the computed prediction depends on the engineer's understanding of the site making this first step an important step in any numerical analysis.

1.5 Scenarios

Once the field conditions are established, it is necessary to build a mechanical understanding of the problem and to define the objectives of the numerical analysis. It is rarely possible to answer all the questions with a single analysis and a list of potential scenarios must be generated. This permits defining which mechanism should be simulated and establish the limitation of the analysis. In some cases, several mechanisms might be of interest in which case several

series of simulations might be required. It is also important to have a certain idea of the expected outcome of the numerical simulation as this facilitates the validation of the model and the debugging process.

It is necessary at this stage to define how the simulations will be presented and which elements and variables are of interest for the analysis so that the modelling efforts go into obtaining these deliverables. This could be the internal forces of a tunnel lining, in which case the lining can be modelled with beam elements rather than a continuum material. It could also be the development of the shear strains in a slope in order to identify the surface of rupture. In some cases, elements or material points (MPs) may need to be monitored during the computations and the desired information recorded in a file, as is demonstrated in [Chapter 12](#).

1.6 Idealisation

It is rarely possible to perform an analysis in which the full knowledge and detail of the object of interest permits a complete, detailed and accurate prediction. Not only is it rarely possible, it is rarely desirable as the understanding is enhanced by intelligent simplifications and idealisation with which the engineer can gain insight [207]. Moreover, it is difficult to create a good model that enables a realistic analysis of the physical processes involved in a real project and that provides a realistic prediction of design quantities [45]. A certain level of idealisation and interpretation is required.

1.6.1 Computational methods

Once the object of interest and the prevailing mechanisms are defined, a computational method needs to be selected. This choice is driven by the technical capacity of the method to capture the correct mechanics and to produce the desired deliverables. The choice of the method is also dependent on the availability of the method (software) and the available computational resources as some methods are computationally expensive. The choice of the numerical methods influences the idealisation of the problem.

A plethora of different computational methods exists ranging from the simplest (e.g. empirical methods) to more advanced methods (e.g. FEM or MPM). Each method has its advantages and limitations. Soga et al. [278] reviewed the different methods for large deformation simulations and a few families of continuum-mechanics methods are summarised as follows.

The empirical methods (EMs) are based on past experiences and theoretical developments. They are often simple enough to be computed by hand or with a spreadsheet. However, EMs rarely consider interactions

between different entities of the system and they are often not adequate for large deformation analyses. They rely on simple mechanical idealisation of the material. Nevertheless, they provide a simple tool for basic predictions and can be used as a verification tool for more advanced numerical predictions. EMs can produce conservative predictions in comparison with more advanced numerical methods [230].

The limit equilibrium methods (LEMs) usually decompose the problem into slices or blocks and check whether the ground is in a state of equilibrium. This is achieved by solving Newton's governing equations and imposing a failure criterion. The use of simple failure criterion implies that the failure is determined from the stress state only and neglects the stress path to failure, and that the criterion holds everywhere along the failure surface [49] but the resulting stresses do not necessarily satisfy equilibrium at every point [276]. The failure surface has to be defined by the user, which often requires many trials in order to determine the critical case. These methods also idealise the drainage conditions as drained or undrained, excluding partially drained conditions from the analysis [230]. LEM is computationally cheap and it can easily be carried out by hand or with a spreadsheet, even for a large number of cases. Fig. 1.2 (a) shows an example of the LEM applied to a strip footing.

The EMs and LEMs separate deformation (serviceability) from strength (failure) and struggle to include any form of interaction in the analysis. However, the long history of these methods resulted in widely available tools and extensive collective experience concerning their reliability [263, 276]. Chapter 11 discusses the use of LEMs and MPM.

The limit analysis methods (LAMs) are based on the plastic bounding theorems and normality (associated flow rule, see Chapter 8). The lower-bound theorem seeks for admissible stress fields such that equilibrium is satisfied [276] but not the compatibility conditions [233]. It is possible to have different stress fields that satisfy the same conditions. The upper-bound theorem seeks the kinematically admissible velocity field satisfying the velocity boundary conditions [276]. It reaches a solution in absence of equilibrium giving unsafe estimates [233]. Human estimate will always be an upper bound as soil will always find cleverer modes of failure [207]. Fig. 1.2 (b-c) shows an example of the LAM applied to a strip footing.

The limit theorems of plasticity provide more flexible and comprehensive solutions because the upper bound introduces a kinematic compatibility consistent with the hypothesis of associated failure criterion, which is absent in LEM. The combination of LAM with FEM (e.g. [276]) and optimisation algorithms has resulted in fast computer programmes for stability analysis.

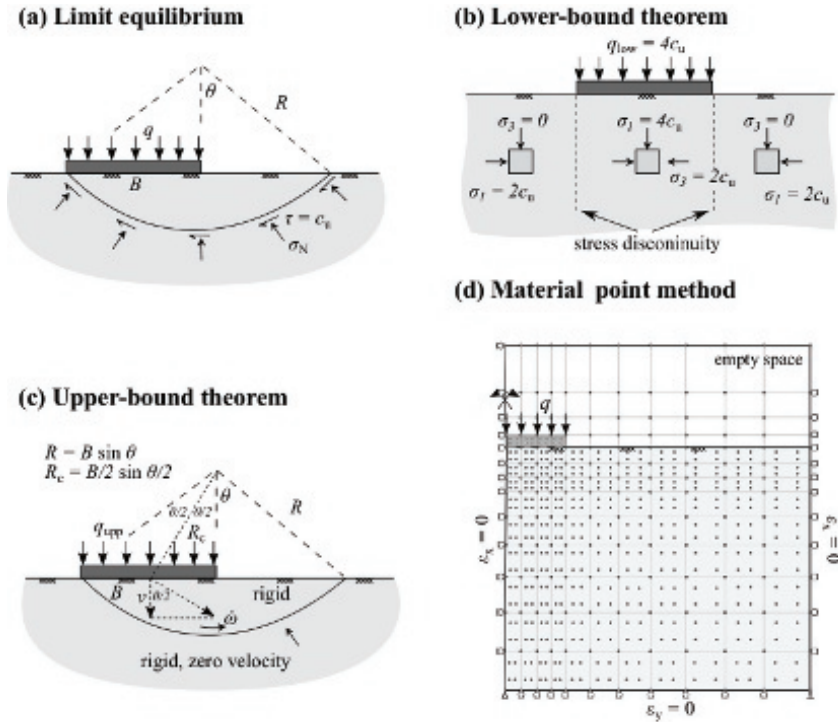


FIGURE 1.2: Schematic description of (a) the limit equilibrium method, (b-c) the limit analysis method and (d) the material point method of a strip footing (original example from [276]).

The finite element method (FEM) solves Newton’s governing equations in order to obtain the stress-strain states with respect to loading, and discretises the continuum into non-overlapping components. It also allows the use of complex soil models. FEM is the most popular numerical method in geotechnical engineering and it is well documented (e.g. [162, 231, 232]). However, FEM struggles with large deformation simulations due to mesh distortion problems. However, an advanced algorithm such as the arbitrary Lagrangian-Eulerian FEM (ALE FEM) offers freedom in changing the computational mesh but it requires a significant computational effort in some cases.

The finite difference method (FDM) approximates derivatives in the partial differential equation by linear combinations of function values at the grid points. It is capable of simulating large deformation problems by updating the coordinates of the grid. It also uses prismatic elements which makes the modelling of complex geometries difficult.

The material point method (MPM) is an Eulerian-Lagrangian method, which discretises the continuum medium in material mass points. It moves these points in a computational grid by solving Newton's governing equations. MPM does not suffer from mesh distortion and it has the advantage of being able to solve the governing equations for the different phases with distinct layers of MPs [21, 334]. However, MPM is computationally expensive and suffers from numerical instabilities related to MPs crossing cells. Chapter 2 presents the fundamental theory of MPM, and Chapters 3 and 4 the different MPM formulations. Fig. 1.2 (d) shows an example of the MPM applied to a strip footing.

The smoothed particle hydrodynamics (SPH) method idealises the continuum material with spaced particles over which the particle properties are smoothed by a kernel function. It is suitable for large deformation simulations but it is computationally expensive, partly because the boundary conditions have to be modelled with a series of particles. Chapter 21 compares the use of SPH and MPM for hydraulic engineering.

1.6.2 Types of analysis

Different types of analysis can be carried out depending on the nature of the problem being solved and the desired deliverables. A few types of simulations are as follows.

Drained, undrained and partially drained analyses define how the excess pore water pressure is dissipated in the simulation, which plays a central role in the mechanical behaviour of soils and rocks.

- A *drained analysis* assumes the absence of excess pore water pressure because of the absence of water (dry soil) or because the deformation is slow enough that the excess pore pressure is rapidly dissipated. It thus depends on the hydraulic conductivity of the soil. Chapter 9 presents simulations of granular column collapses for dry sand and Chapter 15 of pile installations in saturated drained conditions.
- An *undrained analysis* assumes that the deformation is fast and excess pore pressure is accumulated in the ground. There is a high potential for error in undrained simulations as there is not a single constitutive model which can reliably predict the undrained effective stress path, although some models predict more realistic behaviours than others. For this reason, total stress analyses are sometimes favoured as the total stress paths are known and predictable. In undrained effective stress analyses, the effective parameters are used in the constitutive model and the bulk modulus of the water is added to the stiffness of the soil to make the ground incompressible. Excess pore pressure can be computed in effective stress analyses. It is possible to carry

out an undrained total stress analysis for which the total stress paths are correct but for which it is not possible to obtain the excess pore pressure. In such simulations, the total stress parameters are used for the constitutive model and these differ from the effective stress parameters. Total stress analyses are usually carried out for situations in which the undrained response is of interest and a single undrained shear strength can be assigned to the soil element [207].

Despite the difficulty in carrying out undrained analyses, these are often the conditions in which failures take place. Many chapters of this book present undrained analysis using both the total stress and the effective stress approach. For instance, [Chapter 11](#) presents undrained effective stress simulations with a simple constitutive model, and [Chapters 10](#) and [18](#) with an advanced constitutive model. [Chapters 13](#) and [15](#) show examples of total stress analyses.

- A *partially drained or consolidation analysis* considers the dissipation of excess pore pressure with time. It is thus an effective stress analysis. This permits transitioning from undrained conditions to drained conditions. This drainage condition can only be investigated by means of numerical analyses [230]. [Chapters 18](#) and [20](#) show examples of partially drained analyses.

Strength-reduction analyses are the process of progressively reducing the strength of the soil until a failure occurs. The ratio between the strength at failure and the initial strength of the soil gives a factor of safety. This kind of analysis permits identifying the critical strength required for the soil to be stable. An example of strength-reduction analysis is given in [Chapter 15](#).

Inverse analyses are used to back-calculate model parameters such that the simulation will match the field data. The danger with such analyses is that the model parameters become a set of random variables in which only one set of values is obtained without considering mechanical consistency and for which the model parameters may lose their physical meaning. Therefore, the back-calculation of model parameter values must be carried out with caution. An example of inverse analysis is presented in [Chapter 10](#).

Phased analyses are conducted by running a sequence of different types of analysis in an attempt to reproduce field conditions. Nearly all simulations are phased in one way or another as the initial stresses are first generated and then the model is loaded. However, it is possible in some cases to also perform consolidation analysis prior to or after loading, or to carry out a strength-reduction analysis at the end of the simulation in order to obtain a factor of safety.

1.6.3 Model

Once the objectives of the simulations and the type of analysis are defined, the model itself can be generated.

The geometry of the model has to be defined. The components of the model need to be simplified sufficiently for computational reasons and to facilitate the understanding of the problem. However, it also needs to capture enough details in order to predict the correct mechanics. For instance, the model stratigraphy of the ground has to be defined such that it reflects the real stratigraphy. The dimensions of the model have to be large enough to include items of interest and to prevent any influence of the boundary conditions.

The configuration of the model has to be determined and it is a function of the symmetry of the problem. It can be (1) plane strain, (2) axisymmetric or (3) three-dimensional (3D).

- *Plane strain models* are two-dimensional idealisations of the object of interest. They assume a nil deformation in the out-of-plane direction ($\varepsilon_z = 0$) and, hence, the shear stresses can only be non-nil in the plane of analysis. This assumption is representative of long objects such as tunnels, dykes, strip foundations and some excavation. It is not rare to simplify the models according the symmetry of the initial configuration. However, this simplification is only possible if the final configuration is also symmetric. This condition is important for large deformation simulation in which the final configuration can differ substantially from the initial configuration. Fig. 1.3 shows a 2D plane strain idealisation of a dam. Whilst the initial configuration of the dam is symmetric, the pressure of the reservoir on the dam is not. Hence, the model cannot be further simplified with the symmetry of the dam. Additionally, the final configuration of dam is not symmetric.

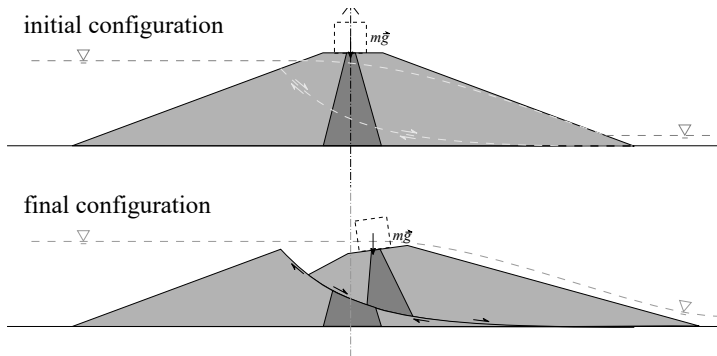


FIGURE 1.3: Initial and final configurations of a numerical model.

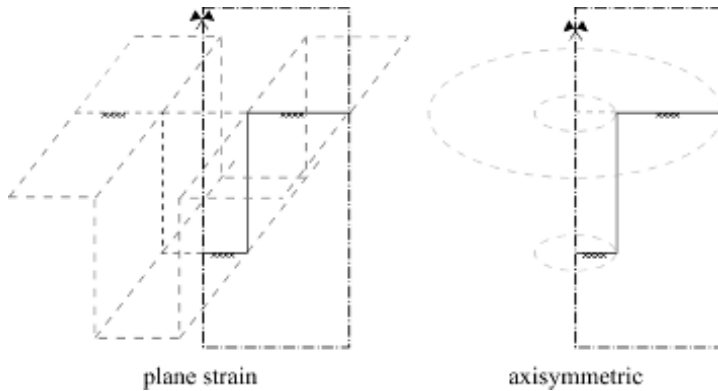


FIGURE 1.4: Two-dimensional idealisation of numerical models.

- *Axisymmetric models* are 2D models, which idealise the object of interest according to a symmetry axis and they assume that the radial strains are nil ($\varepsilon_\theta = 0$). Hence, the shear stresses can only be non-nil in the plane of analysis. In order to carry out an axisymmetric analysis, the initial and final configurations and all loads must be axisymmetric. Fig. 1.4 shows the same idealised excavation in plane strain and axisymmetric conditions but represents two different problems. The plane strain is an idealisation of a trench and the axisymmetric of a shaft. Chapter 5 gives more details on the axisymmetric formulation, whereas Chapter 15 shows examples of plane strain models of excavations.
- *3D models* do not make any assumptions on the out-of-plane direction. However, these simulations require significantly more computational resources as well as being more difficult to analyse. Therefore, 3D simulations are only carried out when 2D models cannot be used. Chapter 19 gives an example of a 3D simulation for which a 2D model could not be used.

The mesh is the discretisation of the space into elements. The mesh is sometimes referred to as the background or computational grid as it is only used for computation reasons and it is not dependent on the geometry of the objects. The elements can also be referred to as cells. However, the mesh must follow the shape of the boundary conditions in MPM. The continuum medium is discretised with MPs and this can be done independently from the mesh, although some codes (i.e. Anura3D [14]) use the background grid to discretise the continuum bodies into MPs. Thus, the size of the mesh defines the amount of mass that a given MP holds. There is no ubiquitous set of rules on how a mesh should be defined as this depends on the problem being solved. However, a good mesh should be as regular as possible with smooth transition between different element sizes. Elon-

gated elements should be avoided as much as possible as it influences the accuracy of the computation. The boundaries of the mesh should be far enough to avoid any influence on the area of interest.

Whilst the mesh deforms with the material in FE simulations, the background computational grid in MPM is generally fixed. It is possible to make this grid move in order to reduce numerical instabilities related to MPs crossing into another cell. Examples of the moving mesh are given in [Chapters 17 and 18](#).

Different element types can be used to model the different components of the model. The ground is typically modelled with continuum elements. Some thin layers or discontinuities can also be modelled as interface elements in FEM or as contact conditions in MPM (see [Chapter 6](#)). Structural elements are typically modelled as beam, plate, struts or anchor elements. However, it may be interesting to model them as a continuum in other cases. This depends on the desired output type as well as their thickness.

The boundary conditions have to be defined such that they mimic the field conditions. Different types of boundary conditions exist. The most common are displacement boundary conditions, which constrain displacements in a given direction by imposing a nil velocity field. It is also possible to assign pressure conditions, which are necessary when modelling unsaturated soil (see [Chapter 12](#)). Conditions can also be imposed within the model and for part of the simulation. This is typically the case when initialising the stresses. [Chapter 9](#) shows an example of such a technique to model the granular column collapse. Inflow and outflow conditions can be imposed in order to model liquid flows. These conditions permit the creation and destruction of MPs during the simulation and an example is given in [Chapter 21](#). It is also possible to remove MPs in a specific area in order to model excavations (see [Chapter 15](#)). The removal of MPs can be spontaneous replicating an excavation and, after that, neighbouring MPs can move into the excavated zone. The removal of MPs can also be applied continuously, removing any MP moving into the predefined area. More information on the boundary conditions is given in [Chapter 6](#).

Constitutive models are mathematical idealisations of the material behaviour, which characterise the stress-strain relationship and define the stiffness of the material. However, there is not a single constitutive model which can adequately predict all the mechanical behaviours of a real soil. A few examples of differences between real and remoulded soil are related to soil structure, homogeneity, anisotropy, beddings and discontinuities. Burland [48] pointed out that, although the constitutive models are largely developed for remoulded soils, they still offer tremendous insight in the mechanical behaviour of real soil.

Constitutive models have evolved significantly throughout the history of geomechanics and Potts et al. [229] classifies them as follows.

- The *first generation* of models are simple elastic-plastic models (Fig. 1.5 a). The behaviour is assumed to be elastic until a failure occurs, after which plastic deformation takes place. The elastic phase can be linear (e.g. [139]) or non-linear (e.g. [90]). The failure threshold is defined by a failure criterion and it is solely predicted by the stress state irrespective of the stress path. Different failure criteria exist such as Mohr-Coulomb [296], Matsuoka-Nakai [197], Drucker-Prager [86] or Hoek-Brown [133, 135]. These models are simple enough for hand calculations but they are also popular in numerical predictions because of their simplicity in understanding and computational efficiency.
- The *second generation* of models are incremental stress-strain relationships (Fig. 1.5 b). These models consider the stress-path dependency to failure and permit the inclusion of plasticity in the hardening phase. Moreover, they are based on mechanical theories which impose a mechanical consistency between stresses and strains. This is not the case for first generation models. Such models are the Original and Modified Cam-Clay models [248, 249], which are presented in Chapter 8.
- The *third generation* of models are the latest generation of models (Fig. 1.5 c). They include extended hardening concepts, bounding surface or multi-surface plasticity, and bubble models. Many of these models are extensions of the Cam-Clay models such as Nor-Sand [148] (see Chapter 8) or Cam-Clay with sub-loading surface [124].

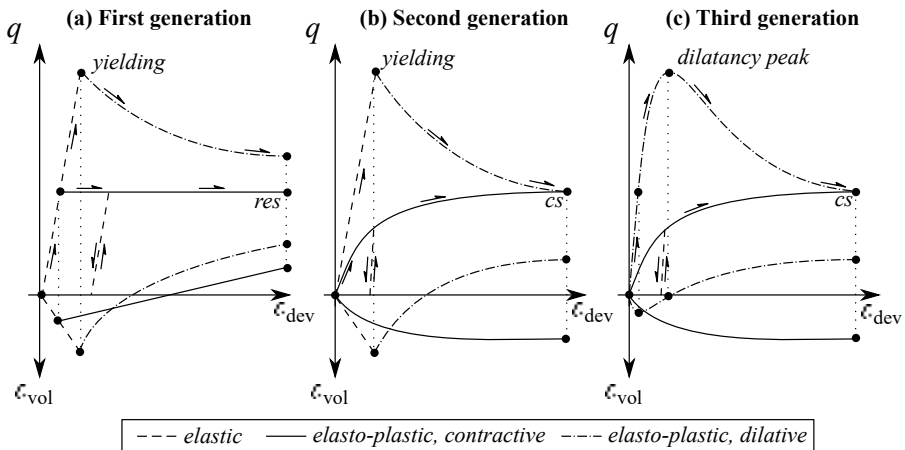


FIGURE 1.5: Schematic description of first, second and third generation of constitutive models for drained triaxial compression test.

A constitutive model is chosen such that it captures at best the mechanics necessary for the prediction. However, it must be compatible with the computational method [232] and the loading sequence. For instance, the ability to reproduce the cyclic loading behaviours is necessary for earthquake engineering, whereas the ability to model unloading paths is necessary for excavation problems. The choice of constitutive model also depends on the available data from which the model parameters are calculated.

The accuracy of a numerical prediction depends heavily on the constitutive model. This does not imply that complex models need to be used. For instance, the bedrock is often modelled as linear elastic when the object of interest is located well above it. However, the selection of a constitutive model requires a thorough understanding of the mechanics of interests, and extensive knowledge of computational geomechanics. Some of the most popular constitutive models can give non-sensible predictions, which is a serious pitfall for practice [230].

Loads have to be defined in the model and this can be done in different ways. The compatibility conditions of MPM impose that the loads are applied at the nodes in contact with a continuum material or a structural element. However, the large deformation nature of MPM allows MPs to move from one element to another and hence MPs may no longer be subjected to this load. It is thus possible to transfer this load from a given node to another one as MPs move (Fig. 1.6). It is also possible to introduce loads in the model as a pressure condition as is the case for gas and liquid pressures. An alternative loading scheme is to model a fictitious continuum body with its weight acting as the load. The advantage of such an approach is that this load can move within the model as the soil deforms (Fig. 1.3).

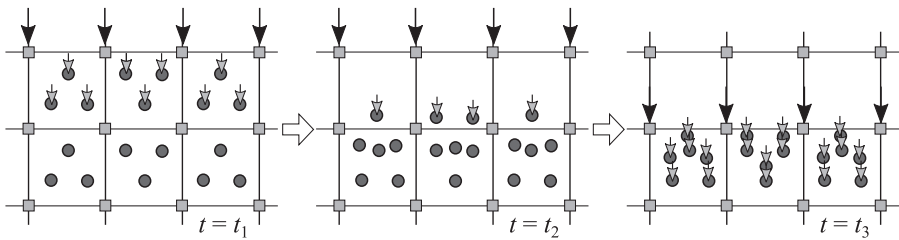


FIGURE 1.6: Schematic description of loads applied on material points.

1.7 Parameters

The quantification of parameters is a difficult step in the analysis as it strongly depends on the type of analysis, constitutive model and model specificities, which include numerical damping of the system. There are considerable discussions on which parameter values to use in a simulation. This is partly due to the lack of physical meaning of some parameters or an inability to quantify the model parameters from experimental data. Two families of parameters can be defined – mechanical and numerical parameters.

Mechanical parameters are parameters used to predict the mechanical behaviour, typically for the constitutive model, but it can also be for contact conditions. The quantification of these parameters depends on the choice of the model and the available data. Mechanical parameters can sometimes be estimated from site and laboratory tests providing that these parameters have physical meanings and are quantifiable. However, one should make sure that the laboratory and site tests are representative of the field conditions. It is well known in rock mechanics that there is a significant difference between the mechanical behaviour of intact rock and rock mass. Soils can also contain some large scale fissures which alter the mechanical behaviour of the soil mass [49]. Sampling techniques can disturb the specimen and, thus, the laboratory test no longer reflects the field conditions (e.g. [69, 164]). Therefore, the determination of the model parameters requires an in-depth knowledge of geomechanics, constitutive modelling and laboratory testing. Potts and Zdravković [232], Lees [162] and Jefferies and Been [150] provide additional information on how parameters can be derived from laboratory tests.

In absence of data, it is sometimes possible to derive model parameters from empirical or theoretical relationships. In some rare cases, catalogues of material characteristics are provided in which parameters are given. However, the use of catalogued material parameters can be misleading as the material is often site specific and state dependent.

Numerical parameters are used to solve the governing equations and they are as important as mechanical parameters. However, the absence of physical meaning makes them more difficult to quantify. A few numerical parameters are described below.

The *characteristic element length* defines the size of an element. However, it does not control the mesh quality. The size of elements plays an important role in the analysis as the solution is mesh-size dependent. Issues related to the size of the mesh are presented in [Chapters 9 and 19](#).

The *damping coefficient* reduces the out-of-balance force in an attempt to reduce numerical oscillations. It is also used to reduce the amount of deformation taking place in the ground. This is often done during the

gravity loading phase. [Chapter 6](#) gives more information on different types of damping and typical values of the damping coefficient.

The *Courant number* sizes the critical time step and, hence, influences the computational time and accuracy (see [Chapter 3](#)). Small Courant numbers give small critical time steps but increase the computational time. Small time steps generally improve the accuracy of the predictions but very small time steps can also decrease the accuracy for thermo-hydro-mechanical coupled problems.

The *number of load steps* and the *load step duration* divide the simulations into a series of steps (see [Chapter 2](#)). Loading conditions, material and numerical parameters and type of analysis can change at the beginning of each time step and the results are saved at the end of each steps. The step duration can change throughout the simulation in order to reduce the size of the output files and focus on the moment of interest. This could be the onset of failure and post-failure behaviour of a dyke failure.

Tolerances and substepping are used as criteria for convergence and these are used throughout the codes. Small values of tolerance generally increase the accuracy of the prediction but they can result in a substantial increase in the computational time. For instance, most constitutive models use substepping techniques in order to guarantee the accuracy of the prediction and for which a tolerance is given and an error is calculated (see [\[275\]](#) for more information). However, errors can be immense when stresses are very low ($\sigma' \rightarrow 0$). The error is calculated by dividing a very small number by another very small number, which can give a big error. This causes unnecessary substepping and computational cost. This is the case for the free surface of a granular flow.

A more in-depth discussion on the numerical parameters is given in [Chapters 3 and 6](#).

1.8 Verification and validation

Verification and validation are two important steps in the analysis [\[45\]](#).

The verification of a prediction consists of making sure that all conditions and parameters of the model are correctly input in the computational code, that the code is performing as expected, and that the sequence of events (i.e. phased analysis) is correctly set. It is making sure that the computer model reflects the mathematical idealisation.

The validation of the prediction consists of making sure that the numerical predictions are representative of the real problem. This is often done comparing the numerical predictions with experimental data, design charts, simplified models, or past experiences

Constitutive models are often validated by simulating laboratory tests with a single stress-point simulation, which provides a certain level of confidence. However, the ability to replicate a given test with a single stress-point simulation does not guarantee that the correct model parameters are used and that the observed behaviour in the single stress-point simulation will be the same in a larger model.

Model boundaries are introduced to limit the extent of the model and reduce the computational time. However, these model boundaries can influence the results as they impose a given stress or strain condition. They can also cause undesired reflections of waves as is demonstrated in [Chapter 20](#). The verification of the boundary influence is done by carrying out a series of simplified model simulations with different dimensions or by running the same simulations with different mesh sizes and comparing the results.

Initial conditions have to be verified after the stress initialisation. This is important as soil is stress-path and pressures dependent. In undrained analyses, it is essential to recreate the pore water pressure distribution, which might require a separate groundwater flow calculation to be performed.

It is important to be aware of the limitations and assumptions of a given simulations as well as some standard pitfalls such as the effect of strain softening or dilative soil response in undrained conditions, which can lead to high effective stresses and soil strength [229]. Locking effects, which are mismatches between the kinematic deformation possibilities and flow rules, can also lead to unrealistic overshoots of stresses.

1.9 Predictions and analysis

Once the modelling process is completed and the results are obtained, the results can be interpreted in order to give the numerical prediction an engineering value. The translation from computer model results to engineering meanings can lead to some discrepancies and misinterpretations in addition to the uncertainties and assumptions of the model. Brinkgreve and Engin [45] point out that the most common misinterpretations are related to the interpretation of factors of safety, the behaviour of structural elements, overlooking of essential details (especially for 3D models), and the general lack of knowledge and understanding of the modelling assumptions.

The translation from numerical predictions to engineering meanings can be straightforward but it can also be more challenging. For instance, the computations of factor of safety is difficult and the criterion of failure has to be clearly defined (see [Chapter 11](#)).

Portraying a prediction of the most probable outcome is often the interpretation of a series of simulations in which different scenarios are investigated. Results are often presented in terms of contour plots of a given variable. Whilst this type of deliverable permits building an understanding, it can be difficult to extract important information and the values for engineering. The preliminary determination of a variable or object of interest, when designing the numerical model, facilitates predictions and discussions.

1.10 Closure

This chapter explains the process of carrying out numerical predictions. It describes each step of the process as well as introducing different types of simulations. Whilst the chapter does not provide the scientific details of numerical analyses (which is the matter of the following chapters), it offers a broad picture of what numerical predictions are and the different levels of idealisation which have to be made. It is shown that the initial state of knowledge of a given problem prior to the analysis influences the numerical model and, hence, the prediction.

As a rule of thumb, numerical models and analyses should go from the simplest ones to more complicated ones as numerous levels of idealisation are required. This approach allows the validation in a step-wise process in order to finally obtain a robust model and prediction. It is important to keep in mind that dummy parameter values might permit the computation to run but they often lead to incorrect predictions creating an illusion of reality. It is important to challenge the model by running a series of tests that verify the model is correctly setup. Finally, it is important to keep in mind that numerical predictions are by definition an idealised interpretation of reality.

References

- NEN 99971. Geotechnical Design - part 1: General Rules. Technical report, Nederlands Normalisatie Instituut, Delft, The Netherlands, 2011.
- Aaron J , Hungr O , Stark T , Baghdady AK . Oso, Washington, Landslide of March 22, 2014: dynamic analysis. Journal of Geotechnical and Geoenvironmental Engineering. 2017;143(9):110.
- Abe K , Soga K , Bandara S . Material point method for coupled hydromechanical problems. Journal of Geotechnical and Geoenvironmental Engineering, page. 2013;04013033.
- K. Abe, K. Soga, and S. Bandara. Material point method for hydro-mechanical problems and its application to seepage failure analyses. In 10th World Congress on Computational Mechanics, pages 3041, 2014.
- I.K.J. Al-Kafaji. Formulation of a dynamic material point method (MPM) for geomechanical problems. PhD thesis, University of Stuttgart, 2013.
- Alonso EE , Gens A , Lloret A . The landslide of Cortes de Pallas. Spain. Gotechnique. 1993;43(4):507521.
- Alonso EE , Pinyol NM . Criteria for rapid sliding I. A review of Vaiont case. Engineering Geology. 2010;114(34):198210.
- E.E. Alonso and N.M. Pinyol. The Vaiont landslide revisited. In Proceeding 2nd Ital. Work. Landslide, pages 924, Napoli, 2012.
- Alonso EE , Pinyol NM , Puzrin AM . Geomechanics of Failures. Advanced Topics: Springer; 2010.
- Alonso EE , Zervos A , Pinyol NM . Thermo-poro-mechanical analysis of landslides: from creeping behaviour to catastrophic failure. Gotechnique. 2016;66(3):202219.
- Alshibli KA , Hasan A . Spatial variation of void ratio and shear band thickness in sand using X-ray computed tomography. Gotechnique. 2008;58(4):249257.
- M. Alvarado. Landslide motion assessment including thermal interaction. An MPM approach. PhD thesis, Universitat Politècnica de Catalunya, 2018.
- 388 Andreykiv A , Keulen F , Rixen DJ , Valstar E . A level-set-based large sliding contact algorithm for easy analysis of implant positioning. International Journal of Numerical Methods in Engineering. 2012;89(10):13171336.
- Anura3D. Anura3D MPM software v. 2017.2. Anura3D MPM Research Community, Delft, 2017.
- Arroyo M , Butlanska J , Gens A , Calvetti F , Jamiolkowski M . Cone penetration tests in a virtual calibration chamber. Gotechnique. 2011;61(6):525531.
- Badger TC . SR 530 MP 35 to 41 Geotechnical Study. Washington State Department of Transportation, Olympia: Technical report; 2015.
- Badger TC . SR 530 Landslide: geotechnical study. Washington State, Department of Transportation, Olympia: Technical report; 2016.
- Baligh MM . Strain path method. Journal of Geotechnical Engineering. 1985;111(9):110836.
- Balmforth NJ , Kerswell RR . Granular collapse in two dimensions. Journal of Fluid Mechanics. 2005;538:399.
- S. Bandara. Material point method to simulate large deformation problems in fluid-saturated granular medium. PhD thesis, University of Cambridge, 2013.
- Bandara S , Soga K . Coupling of soil deformation and pore fluid flow using material point method. Computers and Geotechnics. 2015;63:199214.
- Bardenhagen SG , Brackbill JU , Sulsky DL . The material-point method for granular materials. Computer Methods in Applied Mechanics and Engineering. 2000;187(34):529541.
- Bardenhagen SG , Guilkey JE , Roessig KM , Brackbill JU , Witzel WM , Foster JC . An improved contact algorithm for the material point method and application to stress propagation in granular material. C. - Computational Modelling Engineering Science. 2001;2(4):509522.
- Bardenhagen SG , Kober EM . The generalized interpolation material point method. Computer Modeling in Engineering and Sciences. 2004;5(6):477496.
- Bathe KJ , Ramm E , Wilson EL . Finite element formulations for large deformation dynamics analysis. International Journal of Numerical Methods in Engineering. 1975;9(2):353386.
- 389 Bauer E . Calibration of a comprehensive hypoplastic model for granular materials. Soils and Foundations. 1996;36(1):1326.
- Bear J . Dynamics of Fluids in Porous Media. Dynamics of Fluids in Porous Media: American Elsevier Publishing Company; 1972.
- Been K , Jefferies MG . A state parameter for sands. Gotechnique. 1985;35(2):99112.
- Beget JE . Postglacial volcanic deposits at Glacier Peak, Washington, and potential hazards from future eruptions. USGS: Technical report; 1982.
- L. Beuth. Formulation and application of a quasi-static Material Point Method. PhD thesis, University of Stuttgart, 2012.
- L. Beuth, T. Benz, P.A. Vermeer, C.J. Coetzee, P. Bonnier, and P. van den Berg. Formulation and validation of a quasi-static material point method. In 10th International Symposium in Numerical Methods in GeoMechanics, pages 189195, Rhodes, Greece, 2007.
- Biot MA . General theory of three dimensional consolidation. Journal of Applied Physics. 1941;12:155164.
- Biot MA . Theory of propagation of elastic waves in a fluid saturated porous solid. The Journal of the Acoustical Society of America. 1956;28:168178.
- Bishop AW . Discussion on measurement of shear strength of soils. Gotechnique. 1950;2(2):108116.
- Bishop AW . The use of the slip circle in the stability analysis of slopes. Gotechnique. 1955;5(1):717.
- L. Bjerrum. Embankments on soft ground. In Perform. Earth-Supported Struct., pages 154. Purdue University, 1972.
- Bjerrum L , Eide O . Stability of strutted excavation in clay. Gotechnique. 1956;6(1):3247.
- K. Bock. Rock Mechanics Analyses and Synthesis: RA Experiment. Mont Terri Technical Report 2000G02. Technical report, Q+S Consult, 2001.
- Bolognin M , Martinelli M , Bakker KJ , Jonkman SN . Validation of material point method for soil fluidisation analysis. Journal of Hydrodynamics. 2017;29(3):431437.
- Bolton MD . The strength and dilatancy of sands. Gotechnique. 1986;36(1):6578.
- 390 Boulanger RW . Relating to relative state parameter index. Journal of Geotechnical and Geoenvironmental Engineering. 2003;129(August):770773.
- Brackbill JU , Kothe DB , Ruppel HM . FLIP: A low-dissipation, particle-in-cell method for fluid flow. Computer Physics Communications. 1988;48(1):2538.
- Bray JD , Sancio RB . Assessment of the liquefaction susceptibility of fine-grained soils. Journal of Geotechnical and Geoenvironmental Engineering. 2006;132(9):11651177.
- R.B.J. Brinkgreve and H.L. Bakker. Nonlinear finite-element analysis of safety factors. In 7th International Conference on Comp. Methods and Advances in Geomechanics, pages 11171122, Cairns, 1991.
- R.B.J Brinkgreve and E. Engin. Validation of geotechnical finite element analysis. In 18th International Conference Soil Mechanics and Geotechnical Engineering, pages 677683, Paris, 2013. Presses des Ponts.
- Broere W , Van Tol AF . Modelling the bearing capacity of displacement piles in sand. Institution of Civil Engineers - Geotechnical Engineering. 2006;159(3):195206.

Burghardt J , Brannon R , Guilkey JE . A nonlocal plasticity formulation for the material point method. *Computer Methods in Applied Mechanics and Engineering*. 2012;225228:5564.

Burland JB . On the compressibility and shear strength of natural clays. *Gotechnique*. 1990;40(3):329378.

Burland JB , Chapman T , Skinner H , Brown M . ICE Manual of Geotechnical Engineering. London: ICE Publishing; 2012.

Calvello M , Cuomo S , Ghasemi P . The role of observations in the inverse analysis of landslide propagation. *Computers and Geotechnics*. 2017;92:1121.

Cascini L , Cuomo S , Pastor M , Sorbino G . Modeling of rainfall-induced shallow landslides of the flow-type. *Journal of Geotechnical and Geoenvironmental Engineering*. 2009;136(1):8598.

F. Ceccato. Study of large deformation geomechanical problems with the Material Point Method. PhD thesis, Università Degli Studi Di Padova, 2015.

Ceccato F . Study of flow landslide impact forces on protection structures with the Material Point Method. In *Landslides Eng. Slopes. Exp. Theory Pract.* 12th International Symp. Landslides. 2016;2:615620.

391F. Ceccato, L. Beuth, and P. Simonini. Study of the effect of drainage conditions on cone penetration with the Material Point Method. In *XV Pan-American Conference Soil Mechanics Geotechnical Engineering*, Buenos Aires, 2015.

Ceccato F , Beuth L , Simonini P . Analysis of Piezocone Penetration under Different Drainage Conditions with the Two-Phase Material Point Method. *Journal of Geotechnical and Geoenvironmental Engineering*. 2016;142(12):4016066.

F. Ceccato, L. Beuth, and P. Simonini. Adhesive contact algorithm for MPM and its application to the simulation of cone penetration in clay. In *Procedia Engineering*, pages 182188, 2017.

Ceccato F , Beuth L , Vermeer PA , Simonini P . Two-phase Material Point Method applied to cone penetration for different drainage conditions. In: Soga K , Kumar K , Biscontin G , Kuo M , editors. *International Symp., GeoMechanics from Micro to Macro* London: Taylor & Francis; 2015. p. 965970.

F. Ceccato, L. Beuth, P.A. Vermeer, and P. Simonini. Two-phase Material Point Method applied to the study of cone penetration. *Computers and Geotechnics*, 2016.

F. Ceccato, A. Bisson, and S. Cola. Large displacement numerical study of 3D plate anchors. *European Journal of Environmental and Civil Engineering*, 2017.

Ceccato F , Simonini P . Numerical study of partially drained penetration and pore pressure dissipation in piezocone test. *Acta Geotechnica*. 2016;12(1):195209.

F. Ceccato and P. Simonini. Study of landslide run-out and impact on protection structures with the material point method. In *Interpraevent 2016 - Conference Proceedings*, 2016.

F. Ceccato, A. Yerro, and M. Martinelli. Modelling soil-water interaction with the material point method. Evaluation of single-point and double-point formulations. In *Numerical Methods in Geotechnical Engineering IX, Volume 1: 9th European Conference on Numerical Methods in Geotechnical Engineering (NUMGE 2018)*, pages 351258, 2018.

Chiew Y-M . Mechanics of local scour around submarine pipelines. *Journal of Hydraulic Engineering*. 1990;116(4):515529.

A. Chmelnizkij and J. Grabe. Untersuchungen zur Wellenausbreitung im Boden infolge impulsartiger Beanspruchungen aus Fallmassen und Explosionen. In *Tagungsband zum 12. Hans Lorenz-Symposium 2016 in Berlin*, Veröffentlichungen des Grundbauinstituts der TU Berlin, volume 70, pages 137148, 2016.

392 Chow VT . Open-Channel Hydraulics. New York: McGraw-Hill; 1959.

Church BN . Geology of Hat Creek Basin - summary of field activities. Technical report: B.C. Department of Mines; 1975.

R.V. Churchill. *Operational Mathematics*. McGraw-Hill, 3rd edition, 1972.

Ciabatta M . La dinamica della frana del Vajont. *Giornale di Geologia XXXII*. 1964;I:139154.

Clayton CRI , Hight DW , Hopper RJ . Progressive destructuring of Bothkennar clay: implications for sampling and reconsolidation procedures. *Gotechnique*. 1992;42(2):219239.

Coelho BZ , Rohe A , Soga K . Poroelastic Solid Flow with Material Point Method. *Procedia Engineering*. 2017;175:316323.

Collier N , Pardo D , Dalcin M , Paszynski M , Calo V . The cost of continuity: A study on the performance of isogeometric finite elements using direct solvers. *SIAM Journal on Scientific Computing*. 2013;2(35):767784.

Cortis M , Coombs C , Augarde W , Brown M , Brennan A , Robinson S . Imposition of essential boundary conditions in the material point method. *International Journal for Numerical Methods in Engineering*. 2018;113(1):130152.

C.A. Coulomb. *Essai sur une application des rgles de maximis & minimis quelques problemes de statique relatif l'architecture*. Imprimerie Royale, 1776.

S. Cuomo, P. Ghasemi, M. Calvello, and V. Hoseeinnezhad. Hypoplasticity model and inverse analysis for simulation of triaxial tests. In M. D. M. Fernandes, editor, *9th European Conference on Numerical Methods in Geotechnical Engineering*, pages 307313. Balkema, 2018.

S. Cuomo, M. Pastor, C. Sacco, and L. Cascini. Analisi delle fasi di rottura e post-rottura di frane tipo flusso. In M. D. M. Fernandes, editor, *XXV CNG - AGI - Roma*. AGI, 2014.

Daerr A , Douady S . Sensitivity of granular surface flows to preparation. *Europhys. Lett.* 1999;47:324330.

De Boor C . A Practical Guide to Splines. New York: Applied Mathematical Sciences. Springer; 2001.

de Gast T , Vardon PJ , Hicks MA . Estimating spatial correlations under man-made structures on soft soils. In: Huang J , Fenton GA , Zhang L , Griffiths DV , editors. *Geo-Risk 2017.*, pp Denver: ASCE; 2017. p. 382389.

393Deltares. Slope stability software for soft soil engineering. *D-Geo Stability*, 2016. v16.1.1.

R.P. Denlinger and R.M. Iverson. Flow of variably fluidized granular masses across three-dimensional terrain: 2. numerical predictions and experimental tests. *Journal of Geophysical Research: Solid Earth*, 106(B1):553566, 2001.

C. Detournay and E. Dzik. Nodal Mixed Discretization for tetrahedral elements. In *4th International FLAC Symp. Numer. Model. GeoMechanics*, pages 0702, Madrid, 2006. Itasca Consulting Group Inc.

Dey S . Free overall in open channels: State-of-the-art review. *Flow Measurement and Instrumentation*. 2002;13(56):247264.

Dhakal TR , Zhang DZ . Material point methods applied to one-dimensional shock waves and dual domain material point method with sub-points. *Journal of Computational Physics*. 2016;325:301313.

J. Dijkstra, W. Broere, and A.F. Van Tol. Density changes near an advancing displacement pile in sand. In *2nd BGA International Conference on Foundations (ICOF)*, pages 545554, 2008.

Dodagoudar GR , Sayed S , Rajagopal K . Random field modeling of reinforced retaining walls. *International Journal of Geotechnical Engineering*. 2015;9(3):229238.

Drucker DC , Prager W . Soil mechanics and plastic analysis or limit design. *Quarterly Applied Mathematics*. 1952;10(2):157165.

Drucker DC , Gibson RE , Henkel DJ . Soil mechanics and work-hardening theories of plasticity. *Journal of Soil Mechanics and Foundation Engineering*. 1957;122:338346.

Duff PMD , Gilchrist RD . Correlation of lower Cretaceous coal measures. Ministry of Energy, Mines and Petroleum resources, British Columbia: Peace River Coalfield. Technical report; 1983.

Duncan JM . State of the art: Limit equilibrium and finite-element analysis of slopes. *Journal of Geotechnical Engineering*. 1996;122(7):577596.

Duncan JM , Chang CY . Nonlinear analysis of stress and strain in soils. *Journal of Soil Mechanics and Foundation Division*. 1970;96(5):16291653.

Dykes AP , Bromhead EN . The Vaiont landslide: re-assessment of the evidence leads to rejection of the consensus. *Landslides*. May 2017;118:2018.

394 Eckersley D . Instrumented laboratory flowslides. *Gotechnique*. 1990;40(3):489502.

Eckersley JD . Flowslides in stockpiled coal. *Engineering Geology*. 1985;22(1):1322.

Eingin HK , Brinkgreve RBJ , Van Tol AF . Simplified numerical modelling of pile penetration - the press-replace technique. *International Journal of Numerical and Analytical Methods in Geomechanics*. 2015;39(15):17131734.

Ergun S . Fluid flow through packed column. *Chemical Engineering Progress*. 1952;48:8994.

V. Escario and F. Juca. Strength and deformation of partly saturated soils. In 12th International Conference Soil Mech and Found. Eng., pages 4346, 1989.

Faheem H , Cai F , Ugai K , Hagiwara T . Two-dimensional base stability of excavations in soft soils using FEM. *Computers and Geotechnics*. 2003;30(2):141163.

Farias MM , Naylor DJ . Safety analysis using finite elements. *Computers and Geotechnics*. 1998;22(2):165181.

Fenton GA , Griffiths D , Cavers W . The random finite element method (rfem) in settlement analyses. In: Griffiths DV , Fenton GA , editors. *Probabilistic methods in geotechnical engineering*. CISM courses and lectures 491 Vienna: Springer; 2007. p. 243270.

Fenton GA , Vanmarcke EH . Simulation of random fields via local average subdivision. *Journal of Engineering Mechanics*. 1990;116(8):17331749.

E.J. Fern. Constitutive modelling of unsaturated sand and its application to large deformation modelling. PhD thesis, University of Cambridge, 2016.

E.J. Fern, D.A. de Lange, C. Zwanenburg, J.A.M. Teunissen, A. Rohe, and K. Soga. Experimental and numerical investigations of dyke failures involving soft materials. *Engineering Geology*, 2016.

Fern EJ , Soga K . The role of constitutive models in MPM simulations of granular column collapses. *Acta Geotechnica*. 2016;11(3):659678.

Fern EJ , Soga K . Granular Column Collapse of Wet Sand. *Procedia Engineering*. 2017;175:1420.

395I.M.S. Finnie and M.F. Randolph. Punch-through and liquefaction induced failure of shallow foundations on calcareous sediments. In International Conference on Behaviour of Offshore Structures, Boston, MA, pages 217230, 1994.

Fraccarollo L , Toro EF . Experimental and numerical assessment of the shallow water model for two-dimensional dam-break type problems. *Journal of Hydraulic Research*. 1995;33(6):843864.

Fredlund DG , Morgenstern NR , Widger RA . The shear strength of unsaturated soils. *Canadian Geotechnical Journal*. 1978;15(3):313321.

Gahalaut K , Kraus J , Tomar S . Multigrid methods for isogeometric discretizations. *Computer Methods in Applied Mechanics and Engineering*. 2013;253:413425.

Galavi V , Beuth L , Coelho BZ , Tehrani FS , Hirscher P , Van Tol F . Numerical simulation of pile Installation in saturated sand using Material Point Method. *Procedia Engineering*. 2017;175:7279.

Gavin K , Cadogan D , Tolooiyan A , Casey P . The base resistance of non-displacement piles in sand. part i: field tests. *Institution of Civil Engineers - Geotechnical Engineering*. 2013;166(6):540548.

Genevois R , Tecca P . The Vajont Landslide: State-of-the-Art. *Italian Journal of Engineering Geology and Environment*. 2013;6:1540.

Gens A , Potts DM . Formulation of quasi-axisymmetric boundary value problems for finite element analysis. *Engineering with Computers*. 1984;1(2):144150.

D. Givoli and D. Cohen. Nonreflecting boundary conditions based on Kirchhoff-type formulae, 1995.

Golub G , van Loan C . *Matrix Computations*. Baltimore: The Johns Hopkins University Press; 1996.

J.L. Gonzalez Acosta, P.J. Vardon, and M.A. Hicks. Composite material point method (cmpm) to improve stress recovery for quasi-static problems. *Procedia Engineering*, 175:324331, 2017.

J. Grabe, T. Pucker, and T. Hamann. Numerical simulation of pile installation processes in dry and saturated granular soils. In International Conference on Numerical Methods in Geotechnical Engineering (NUMGE), pages 663668, 2014.

Griffiths D . Computation of bearing capacity factors using finite elements. *Gotechnique*. 1982;32(3):195202.

396 Griffiths DV , Lane PA . Slope stability analysis by finite elements. *Gotechnique*. 1999;49(3):387403.

Griffiths DV , Fenton GA . Probabilistic slope stability analysis by finite elements. *Journal Geotechnical and Geoenvironmental Engineering*. 2004;130(5):507518.

Gudehus G . A visco-hypoplastic constitutive relation for soft soils. *Soils and Foundations*. 2004;44(4):1125.

Hamad F , Stolle D , Vermeer P . Modelling of membranes in the material point method with applications. *International Journal for Numerical and Analytical Methods in Geomechanics*. 2015;39(8):833853.

Hamad F , Wickowski Z , Moormann C . Interaction of fluid-solid-geomembrane by the material point method. *Computers and Geotechnics*. 2017;81:112124.

Hamann T , Qiu G , Grabe J . Application of a coupled Eulerian-Lagrangian approach on pile installation problems under partially drained conditions. *Computers and Geotechnics*. 2015;63:279290.

Hashiguchi K , Chen Z . Elastoplastic constitutive equation of soils with the subloading surface and the rotational hardening. *International Journal for Numerical and Analytical Methods in Geomechanics*. 1998;22(3):197227.

Haugerud RA . Preliminary Interpretation of Pre-2014 Landslide Deposits in the Vicinity of Oso. USGS, Reston (VA): Washington. Technical report; 2014.

A.J. Hendron and F.D. Patton. The Vaiont slide, a geotechnical analysis based on new geologic observations of the failure surface. Vol. I. US Army Corps of Engineers, 2:187188, 1985.

B. Henn, Q. Cao, D.P. Lettenmaier, C.S. Magirl, C. Mass, J.B. Bower, M. St Laurent, Y.X. Mao, and S. Perica. Hydroclimatic Conditions Preceding the March 2014 Oso Landslide. *Journal of Hydrometeorology*, 16(3):12431249, 2015.

Herle I , Gudehus G . Determination of parameters of a hypoplastic constitutive model from properties of grain assemblies. *Mechanics of Cohesive-Frictional Materials*. 1999;4(5):461486.

Hicks MA , Nuttall JD , Chen J . Influence of heterogeneity on 3d slope reliability and failure consequence. *Computers and Geotechnics*. 2014;61:198208.

397 Hicks MA , Samy K . Influence of heterogeneity on undrained clay slope stability. *Quarterly Journal of Engineering Geology and Hydrogeology*. 2002;35(1):4149.

M.A. Hicks and W. Wong. Static liquefaction of loose slopes. In *Numerical Methods in Geomechanics*, pages 13611367. Balkema, 1988.

Hill R . Some basic principles in the mechanics of solids without a natural time. *Journal of the Mechanics and Physics of Solids*. 1959;7(3):209225.

Hoek E . Strength of jointed rock masses. *Gotechnique*. 1983;33(3):187223.

E. Hoek. Rocscience lecture series, Lecture 2: "The Art of Tunnelling", 2015.

Hoek E , Carranza C , Corkum B . Hoek-brown failure criterion - 2002 edition. In: Hammah Reginald , editor. Narms-Tac., pp Toronto: University of Toronto Press; 2002. p. 267273.

Hofreither C , Takacs S , Zulehner W . A robust multigrid method for isogeometric analysis in two dimensions using boundary correction. *Computer Methods in Applied Mechanics and Engineering*. 2017;316:2242.

P. Holscher, A.F. Van Tol, and N.Q. Huy. Rapid pile load tests in the geotechnical centrifuge. In 9th International Conference on Testing and Design Methods for Deep Foundations, Japanese Geotechnical Society, pages 257263, 2012.

Y. Honjo. Challenges in geotechnical reliability based design. In N. Vogt, B. Schuppener, D. Straub, and G. Bru, editors, 3rd International Symposium on Geotechnical Safety and Risk, pages 1127, 2011.

R. Hooke. De Potentia Restitutiva or of Spring. J. Martyn, London, 1678.

Hosseinneshad V , Rafiee M , Ahmadian M , Ameli MT . Species-based quantum particle swarm optimization for economic load dispatch. *International Journal of Electrical Power & Energy Systems*. 2014;63:311322.

Hu W , Chen Z . A multi-mesh MPM for simulating the meshing process of spur gears. *Computers and Structures*. 2003;81(20):19912002.

Hu Y , Randolph MF . A practical numerical approach for large deformation problems in soil. *International Journal for Numerical and Analytical Methods in Geomechanics*. 1998;22(5):327350.

398 Huang P , Zhang X , Ma S , Huang X . Contact algorithms for the material point method in impact and penetration simulation. *International Journal of Numerical Methods in Engineering*. 2011;85(4):498517.

M. Huber. Soil variability and its consequences in geotechnical engineering. PhD thesis, University of Stuttgart, 2013.

Hungr O , Leroueil S , Picarelli L . The Varnes classification of landslide types, an update. *Landslides*. 2014;11(2):167194.

Iverson RM , George DL . Modelling landslide liquefaction, mobility bifurcation and the dynamics of the 2014 Oso disaster. *Gotechnique*. 2016;66(3):175187.

R.M. Iverson, D.L. George, K. Allstadt, M.E. Reid, B.D. Collins, J.W. Vallance, S.P. Schilling, Jonathan W. Godt, C.M. Cannon, C.S. Magirl, R.L. Baum, J.a. Coe, W.H. Schulz, and J.B. Bower. Landslide mobility and hazards: implications of the . Oso disaster. *Earth and Planetary Science Letters*. 2014;412(197208):2015.

Jefferies MG . Nor-Sand: a simple critical state model for sand. *Gotechnique*. 1993;43(1):91103.

Jefferies MG , Been K . Soil Liquefaction - a Critical State Approach. 1st ed. London: Taylor & Francis; 2006.

Jefferies MG , Been K . Soil Liquefaction - a Critical State Approach. 2nd ed. London: CRC Press; 2016.

Jefferies MG , Shuttle DA . Dilatancy in general Cambridge-type models. *Gotechnique*. 2002;52(9):625638.

Jefferies MG , Shuttle DA . On the operating critical friction ratio in general stress states. *Gotechnique*. 2011;61(8):709713.

Keaton JR , Wartman J , Anderson S , Benot J , de la Chappelle J , Gilbert Robert , Montgomery DR , The 22, Oso Landslide, Snohomish County. Washington: Technical report, Geotechnical Extreme Events Reconnaissance; March 2014.

Kim K , Prezzi M , Salgado R , Lee W . Effect of penetration rate on cone penetration resistance in saturated clayey soils. *Journal of Geotechnical and Geoenvironmental Engineering*. 2008;134(8):11421153.

J. Kirstein, J. Grabe, and A. Chmelnizkij. Numerische Berechnungen und messtechnische Begleitung zur Dynamischen Intensivverdichtung. In Tagungsband zur 34. Baugrundtagung 2016 in Bielefeld, pages 297304, 2016.

399 Klotz EU , Coop MR . An investigation of the effect of soil state on the capacity of driven piles in sands. *Gotechnique*. 2001;51(9):733751.

J.J. Kolbuszewski. An experimental study of the maximum and minimum porosities of sands. In 6th International Conference Soil Mechanics Foundation Engineering, pages 158165, Rotterdam, 1948.

D. Kolymbas. A generalized hypoelastic constitutive law. In XI International Conference Soil Mechanics and Foundation Engineering, volume 5, page 2626, San Francisco, 1985. Balkema.

Lajeunesse E , Mangeney-Castelnau A , Vilotte JP . Spreading of a granular mass on a horizontal plane. *Physics of Fluid*. 2004;16(7):23712381.

Lajeunesse E , Monnier JB , Homsy GM . Granular slumping on a horizontal surface. *Physics of Fluid*. 2005;17(10):103302.

Lambe TW . Predictions in soil engineering. *Gotechnique*. 1973;23(2):151202.

Lees A . Geotechnical finite element analysis: a practical guide. London: Institution of Civil Engineers; 2016.

Legendre AM . Nouvelles mthodes pour la dtermination des orbites des comtes. Paris: F. Didot; 1805.

Leroueil S . Compressibility of clays: fundamental and practical aspects. *Journal of Geotechnical Engineering*. 1996;122(7):534543.

Levadoux JN , Baligh MM . Consolidation after undrained piezocone penetration. i: Prediction. *Journal of Geotechnical Engineering*. 1986;112(7):707726.

Li DQ , Xiao T , Cao ZJ , Zhou CB , Zhang LM . Enhancement of random finite element method in reliability analysis and risk assessment of soil slopes using subset simulation. *Landslides*. 2016;13(2):293303.

Li LC , Tang CA , Zhu WC , Liang ZZ . Numerical analysis of slope stability based on the gravity increase method. *Computers and Geotechnics*. 2009;36(7):12461258.

Li YL , Hicks MA , Nuttall JD . Comparative analyses of slope reliability in 3d. *Engineering Geology*. 2015;196:1223.

Li YL , Hicks MA , Vardon PJ . Uncertainty reduction and sampling efficiency in slope designs using 3d conditional random fields. *Computers and Geotechnics*. 2016;79:159172.

400 Liang D . Evaluating shallow water assumptions in dam-break flows. *ICE - Water Management*. 2010;163(5):227237.

Liang D , Falconer RA , Lin Binliang . Comparison between TVD-MacCormack and ADI-type solvers of the shallow water equations. *Advances in Water Resources*. 2006;29(12):18331845.

D. Liang, X. Zhao, and M. Martinelli. MPM simulations of the interaction between water jet and soil bed. In *Procedia Engineering*, volume 175, pages 242249, Delft, the Netherlands, 2017. Elsevier B.V.

Lim LJ , Andreykiv A , Brinkgreve RBJ . On the application of the material point method for offshore foundations. In: Hicks MA , Brinkgreve RBJ , Rohe A , editors. *Numerical Methods in Geotechnical Engineering*, pp Delft: Taylor & Francis; 2014. p. 253258.

Lombardi G . Les ouvrages souterrains du CERN. Publication de la Socit Suisse de Mcanique des Sols et des Roches. 1981;104:4753.

Lpez-Querol S , Fernndez-Merodo JA , Mira P , Pastor M . Numerical modelling of dynamic consolidation on granular soils. *International J. Numer. Anal. Meth. GeoMechanics*. 2008;32(12):14311457.

Love E , Sulsky DL . An unconditionally stable, energy-momentum consistent implementation of the material-point method. *Computer Methods in Applied Mechanics and Engineering*. 2006;195:39033925.

G. Lube, H. Huppert, R.S.J. Sparks, and A. Freundt. Collapses of two-dimensional granular columns. *Phys. Rev. E - Stat. Nonlinear, Soft Matter Phys.*, 72(4):110, 2005.

Lube G , Huppert H , Sparks RSJ , Freundt A . Static and flowing regions in granular collapses down channels. *Physics of Fluid*. 2007;19(4):043301.

Lube G , Huppert H , Sparks RSJ , Hallworth MA . Axisymmetric collapses of granular columns. *Journal of Fluid Mechanics*. 2004;508:175199.

T. Lunne, J. M. Powell, P. K. Robertson, et al. Cone Penetration Testing in Geotechnical Practice. CRC Press, 1999.

Luo H , Baum JD , Lhner R . A discontinuous Galerkin method based on a Taylor basis for the compressible flows on arbitrary grids. *Journal of Computational Physics*. 2008;227(20):88758893.

Luong M , Touati A . Sols grenus sous fortes contraintes. *Revue Franaise de Gotechnique*. 1983;24:5163.

401J. Lysmer and R.L. Kuhlemeyer. Finite dynamic model for infinite media. In ASCE, pages 859877, 1969.

J. Ma, D. Wang, and M.F. Randolph. A new contact algorithm in the material point method for geotechnical simulations. *International Journal for Numerical and Analytical Methods in Geomechanics*, pages 11971210, 2014.

Ma S , Zhang X , Qiu XM . Comparison study of MPM and SPH in modeling hypervelocity impact problems. *International Journal of Impact Engineering*. 2009;36(2):272282.

Mahmoodzadeh H , Randolph MF . Penetrometer testing: effect of partial consolidation on subsequent dissipation response. *Journal of Geotechnical and Geoenvironmental Engineering*. 2014;140(6):04014022.

K. Mahutka, F. Konig, and J. Grabe. Numerical modelling of pile jacking, driving and vibro driving. In *International Conference on Numerical Simulation of Construction Processes in Geotechnical Engineering for Urban Environment*, pages 235246, 2006.

Malkus DS , Hughes TJR . Mixed finite element methods - Reduced and selective integration techniques: A unification of concepts. *Computer Methods in Applied Mechanics and Engineering*. 1978;15(1):6381.

Malvern LE . Introduction to the Mechanics of a Continuous Medium. Englewood Cliffs, New Jersey: Prentice-Hall; 1996.

Mandal S , Maiti R . Semi-quantitative Approaches for Landslide Assessment and Prediction. London: Springer; 2015.

Martin JC , Moyce WJ . An experimental study of the collapse of liquid columns on a rigid horizontal plane. *Philosophical Transactions of Royal Society London, Series A*. 1952;244(882):312324.

Martinelli M . Soil-water interaction with material point method. Deltares, Delft, the Netherlands: Technical report; 2016.

M. Martinelli, S.T. Faraz, and V. Galavi. Analysis of crater development around damaged pipelines using the material point method. *Procedia Engineering*, 175:204211, 2017. 1st International Conference on the Material Point Method (MPM 2017).

M. Martinelli and A. Rohe. Modelling fluidisation and sedimentation using material point method. In *1st Pan-American Congress on Computational Mechanics*, pages 112, 2015.

402 Martinelli M , Rohe A , Soga K . Modeling dike failure using the Material Point Method. *Procedia Engineering*. 2017;175(2016):341348.

Mast CM , Mackenzie-Helnwein P , Arduino P , Miller GR , Shin W . Mitigating kinematic locking in the material point method. *Journal of Computational Physics*. 2012;231(16):53515373.

Matsuoka H , Nakai T . Stress deformation and strength characteristics of soil under three different principal stresses. *Soil Found*. 1974;232:5970.

Mayne PW , Jones JS Jr . Impact stresses during dynamic compaction. *Journal of Geotechnical Engineering*. 1983;109(10):13421346.

Mesri G , Ajlouni M . Engineering properties of fibrous peats. *Journal of Geotechnical and Geoenvironmental Engineering*. 2007;133(7):850866.

Milligan V . Geotechnical aspects of glacial tills. In: Legget RF , editor. *Glacial Till an Inter-disciplinary Study*. Ottawa: Royal Society of Canada; 1976. p. 269290.

Mitchell JK , Soga K . Fundamentals of Soil Behavior. 3rd ed. New York: Wiley & Sons; 2005.

C.O. Mohr. Welche Umstaad Bedingen des Elastizitaetsgrenzen und den Bruch eines material? In *Abhandlingen aus dem Gebiete der Technischen Mechanik*, pages 192240. Ernst und Sohn, Berlin, 3rd edition, 1928.

Monaghan JJ , Kos A . Scott Russells wave generator. *Physics of Fluids*. 2000;12(3):622630.

Monforte L , Carbonell JM , Arroyo M , Gens A . Performance of mixed formulations for the particle finite element method in soil mechanics problems. *Computational Particle Mechanics*. 2017;4(3):269284.

Montes JS . Potential flow solution to 2D transition from mild to steep slope. *Journal of Hydraulic Engineering*. 1994;120(5):601621.

K.F. Mostafa and R.Y. Liang. Numerical modeling of dynamic compaction in cohesive soils. In *Geo-Frontiers 2011*, pages 738747, Reston, VA, 2011. American Society of Civil Engineers.

D. Muir Wood. *Soil Behavior and Critical State Soil Mechanics*. Cambridge University Press, Cambridge, 1990.

D. Muir Wood and K. Belkheir. Strain softening and state parameter for sand modelling. *Gotechnique*, 44(2):335339, 1994.

Mller L . The rock slide in the Vajont Valley. *Rock Mechanics and Engineering Geology*. 1964;2:148212.

403 Murray E , Geddes JD . Uplift of anchor plates in sand. *Journal of Geotechnical Engineering*. 1987;113(3):202215.

Naghibi F , Fenton GA , Griffiths DV . Probabilistic considerations for the design of deep foundations against excessive differential settlement. *Canadian Geotechnical Journal*. 2016;53:11671175.

Nakagawa K , Soga K , Mitchell JK . Observation of biot compressional wave of the second kind in granular soils. *Gotechnique*. 1997;47(1):133147.

Newland PL , Allely BH . Volume changes in drained triaxial tests on granular materials. *Gotechnique*. 1957;7(1):1734.

Niemunis A , Herle I . Hypoplastic model for cohesionless soils with elastic strain range. *Mechanics of Cohesive-frictional Materials: An International Journal on Experiments, Modelling and Computation of Materials and Structures*. 1997;2(4):279299.

Nonveiller E . The Vajont reservoir slope failure. *Engineering Geology*. 1987;24:493512.

Nova R . A constitutive model for soil under monotonic and cyclic loading. In: Pande GN , Zienkiewicz C , editors. *Soil Mechanics - Transient Cyclic Loading*. pp Chichester: Wiley; 1982. p. 343373.

Nova R , Wood DM . A constitutive model for sand in triaxial compression. *International Journal for Numerical and Analytical Methods in Geomechanics*. 1979;3(3):255278.

Oliveira J.R.M.S. , Almeida MSS , Motta HPG , Almeida MCF . Influence of penetration rate on penetrometer resistance. *Journal of Geotechnical and Geoenvironmental Engineering*. 2011;137(7):695703.

Palladino DJ , Peck RB . Slope failures in an overconsolidated clay, Seattle. *Washington. Gotechnique*. 1972;22(4):563595.

Pan JL , Selby AR . Simulation of dynamic compaction of loose granular soils. *Advances in Engineering Software*. 2002;33(710):631640.

Pankow KL , Moore JR , Hale JM , Koper KD , Kubacki T , Whidden KM , McCarter MK . Massive landslide at Utah copper mine generates wealth of geophysical data. *GSA Today*. 2014;24(1):49.

Paronuzzi P , Bolla A . Gravity-induced rock mass damage related to large en masse rockslides: Evidence from Vajont. *Geomorphology*. 2015;234:2853.

404 Pastor M , Blanc T , Haddad B , Petrone S , Morles MS , Dremptic V , Issler D , Crosta GB , Cascini L , Sorbino G , et al . Application of a sph depth-integrated model to landslide run-out analysis. *Landslides*. 2014;11(5):793812.

Phuong NTV , van Tol AF , Elkadi ASK , Rohe A . Numerical investigation of pile installation effects in sand using material point method. *Computers and Geotechnics*. 2016;73:5871.

Pinyol NM , Alonso EE . Fast planar slides. A closed-form thermo-hydro-mechanical solution. *International Journal for Numerical and Analytical Methods in Geomechanics*. 2010;34(1):2752.

Pinyol NM , Alvarado M , Alonso EE , Zabala F . Thermal effects in landslide mobility. *Gotechnique*. 2018;68(6):528545.

Plona TJ . Observation of a second bulk compressional wave in a porous medium at ultrasonic frequencies. *Applied Physics Letters*. 1980;36(4):259261.

E.P. Poeter and M.C. Hill. Documentation of UCODE: A computer code for universal inverse modeling, volume 98. DIANE Publishing, 1998.

Potts D , Axelsson K , Grande L , Schweiger H , Long M . Guidelines For the Use of Advanced Numerical Analysis. London: Thomas Telford; 2002.

Potts DM . Numerical analysis: a virtual dream or practical reality? *Gotechnique*. 2003;53(6):535573.

Potts DM , Zdravkovi L . Finite Element Analysis in Geotechnical Engineering: Application. London: Thomas Telford; 1999.

Potts DM , Zdravkovi L . Finite Element Analysis in Geotechnical Engineering: Theory. London: Thomas Telford Ltd; 2001.

D.M. Potts and L. Zdravkovic. Computer analysis principles in geotechnical engineering. In J.B. Burland, T. Chapman, H. Skinner, and Brown. M., editors, *ICE Manual of Geotechnical Engineering*, Chapter 6, pages 3558. ICE Publishing, London, 2012.

Potyondy JG . Skin friction between various soils and construction materials. *Gotechnique*. 1961;11(4):339353.

Pyles MR , Rogers JD , Bray JD , Skaugset A , Storesund R , Schlieder G , 2014 Oso Landslide - Expert report. June 30, Technical report. King County: Superior Court of Washington for King County; 2016.

405 Qiu G , Henke S , Grabe J . Application of a coupled Eulerian-Lagrangian approach on geomechanical problems involving large deformations. *Computers and Geotechnics*. 2011;38(1):3039.

Rajaratnam N , Muralidhar D . Characteristics of the rectangular free overfall. *Journal of Hydraulic Research*. 1968;6(3):233258.

Randolph M . Science and empiricism in pile foundation design. *Gotechnique*. 2003;53(10):84775.

Randolph MF , Dolwin R , Beck R . Design of driven piles in sand. *Gotechnique*. 1994;44(3):427448.

M.F. Randolph and S. Hope. Effect of cone velocity on cone resistance and excess pore pressures. In *International Symposium on Engineering Practice and Performance of Soft Deposits*, pages 147152. Yodagawa Kogisha Co., Ltd, 2004.

G. Remmerswaal, M.A. Hicks, and P.J. Vardon. Ultimate limit state assessment of dyke reliability using the random material point method. In *4th International Symposium on Computational Geomechanics*, pages 8990, Assisi, 2018.

Replogle JA . Discussion: "End depth at a drop in trapezoidal channels" by M.H. Diskin. *Journal of the Hydraulics Division*. 1962;88(2):161165.

M.F. Riemer, B.D. Collins, T.C. Badger, C. Toth, and Y.C. Yu. Geotechnical soil characterization of intact quaternary deposits forming the March 22, 2014 SR-530 (Oso) Landslide, Snohomish County, Washington: U.S. Geological Survey Open-File Report 20151089. Technical report, USGS, 2015.

Robertson PK . Cone penetration test (cpt)-based soil behaviour type (sbt) classification system - an update. *Canadian Geotechnical Journal*. 2016;53(12):19101927.

Robertson PK , Sully JP , Woeller DJ , Lunne T , Powell JJM , Gillespie DG . Estimating coefficient of consolidation from piezocone tests. *Canadian Geotechnical Journal*. 1992;29(4):539550.

Rogers JD , Pyles MR , Bray JD , Skaugset A , Storesund R , Schlieder G . 2014 Oso Landslide - Expert report January 22, 2016. Superior Court of Washington for King County, King County: Technical report; 2016.

Roscoe KH . The Influence of strains in soil mechanics. *Gotechnique*. 1970;20(2):129170.

406 Roscoe KH , Burland JB . On the generalised stress-strain behaviour of wet clay. In: Heyman J , Leckie FA , editors. *Engineering Plasticity*, pp Cambridge: Cambridge University Press; 1968. p. 535609.

K.H. Roscoe and A.N. Schofield. Mechanical behaviour of an idealised wet clay. In *2nd Eur. Conference Soil Mechanics Found. Eng.*, pages 4754, Wiesbaden, 1963.

Roscoe KH , Schofield AN , Wroth CP . On the yielding of soil. *Gotechnique*. 1958;8:2252.

Ross CS , Shannon EV . The minerals of bentonite and related clays and their physical properties. *J. Am. Ceram. Soc.* 1926;9(2):7796.

Rouse H . Discharge characteristics of the free overfall: Use of crest section as a control provides easy means of measuring discharge. *Civil Engineering ASCE*. 1936;6(4):257260.

P.W. Rowe. The stress-dilatancy relation for static equilibrium of an assembly of particles in contact. In *Proceedings of the Royal Society. Mathematical, Physical and Engineering Sciences*, volume 269, pages 500527. The Royal Society, 1962.

Saad Y . Iterative Methods for Sparse Linear Systems. Philadelphia: SIAM; 2003.

Salgado R , Mitchell J , Jamiolkowski M . Calibration chamber size effects on penetration resistance in sand. *J. Geotech. Geoenviron. Eng.* 1998;124(9):87888.

Salgado R , Prezzi M . Computation of cavity expansion pressure and penetration resistance in sands. *International Journal of Geomechanics*. 2007;7(4):251265.

Sangalli G , Tani M . Isogeometric preconditioners based on fast solvers for the sylvester equation. *SIAM Journal on Scientific Computing*. 2016;6(38):36443671.

Savage WZ , Morrissey MM , Baum RL . Geotechnical Properties for Landslide-Prone Seattle Area Glacial Deposits: Open-File Report 00228. USGS, Denver: Technical report; 2000.

Schneider JA , Lehane BM , Schnaid F . Velocity effects on piezocone measurements in normally and over consolidated clays. *International Journal of Physical Modelling in Geotechnics*. 2007;7(2):2334.

Schobeiri MT . Fluid Mechanics for Engineers. Berlin: Springer; 2010.

407 Schofield AN . Cambridge geotechnical centrifuge operations. *Gotechnique*. 1980;30(3):227268.

Schofield AN , Wroth P . Critical State Soil Mechanics. 2nd ed. London: McGraw-Hill; 1968.

Schweiger HF . An introduction to numerical modelling in geotechnical engineering. ISSMGE: Technical report; 2017.

Scott RA , Pearce RW . Soil compaction by impact. *Gotechnique*. 1975;25(1):1930.

E. Semenza. La Storia del Vaiont Raccontata del Geologo che ha Scoperto la Frana. Tecomproject Editore Multimediale, 2001.

Semenza E , Ghirotti M . History of the 1963 Vaiont slide: the importance of geological factors. *Bulletin of Engineering Geology and the Environment*. 2000;59:8797.

Shannon WD . Slide on North Fork Stillaguanish River near Hazel. Technical report: William D. Shannon & Associates, Washington; 1952.

Shao S , Lo EYM . Incompressible SPH method for simulating Newtonian and non-Newtonian flows with a free surface. *Advances in Water Resources*. 2003;26(7):787800.

Silva MF , White DJ , Bolton MD . An analytical study of the effect of penetration rate on piezocone tests in clay. *International journal for numerical and analytical methods in geomechanics*. 2006;30(6):501527.

Skempton AW . The bearing capacity of clays. London: In *Building Research Congress*; 1951. p. 180189.

A.W. Skempton. The colloidal activity of clays. In *3rd International Conference Soil Mechanics Found. Eng.*, pages 5761, Zurich, 1953. ICOSOMEF.

Skempton AW . Residual strength of clays in landslides, folded strata and the laboratory. *Gotechnique*. 1985;35(1):318.

Skempton AW , Bishop AW . The measurement of the shear strength of soils. *Gotechnique*. 1950;2(2):90108.

Sloan S , Randolph MF . Numerical prediction of collapse loads using finite element methods. *International Journal of Numerical and Analytical Methods in Geomechanics*. 1982;6(1):4776.

Sloan SW . Substepping schemes for the numerical integration of elastoplastic stress-strain relations. *International Journal of Numerical Methods in Engineering*. 1987;24(5):893911.

408 Sloan SW . Geotechnical stability analysis. *Gotechnique*. 2013;63(7):531571.

J.I. Sneddon. A study of two soils derived from volcanic ash in southwestern British Columbia and a review and determination of ash distribution in western Canada. PhD thesis, University of British Columbia, 1970.

Soga K , Alonso EE , Yerro A , Kumar K , Bandara S . Trends in large-deformation analysis of landslide mass movements with particular emphasis on the material point method. *Gotechnique*. 2016;66(3):248273.

Srivastava A , Babu GLS , Haldar S . Influence of spatial variability of permeability property on steady state seepage flow and slope stability analysis. *Engineering Geology*. 2010;110:93101.

Stark T . Ahmed K. Baghdady, O. Hungr, and J. Aaron. Case study: Oso, Washington, Landslide of March 22, 2014 - material properties and failure mechanism. *Journal of Geotechnical and Geoenvironmental Engineering*. 2017;143(5):05017001.

Stark T , Eid H . Drained residual strength of cohesive soils. *Journal of Geotechnical Engineering*. 1994;120(5):856871.

Steffen M , Kirby RM , Berzins M . Analysis and reduction of quadrature errors in the material point method (MPM). *International Journal for Numerical Methods in Engineering*. 2008;76(6):922948.

Steffen M , Kirby RM , Berzins M . Decoupling and balancing of space and time errors in the material point method (MPM). *International Journal for Numerical Methods in Engineering*. 2010;82(10):12071243.

Sulsky D , Schreyer HL . Axisymmetric form of the material point method with applications to upsetting and Taylor impact problems. *Computer Methods in Applied Mechanics and Engineering*. 1996;139:409429.

Sulsky DL , Chen Z , Schreyer HL . A particle method for history-dependent materials. *Computer Methods in Applied Mechanics and Engineering*. 1994;118:179196.

Sulsky DL , Gong M . Improving the material-point method. In: Weinberg K , Pandolfi A , editors. *Innovative Numerical Approaches for Multi-Fields and Multi-Scale Problems.*, pp Berlin: Springer; 2016. p. 217240.

409 Sulsky DL , Zhou S-J , Schreyer HL . Application of a particle-in-cell method to solid mechanics. *Journal of Computational Physics Community*. 1995;87:236252.

B.M. Sumer and J. Fredse. *The Mechanics of Scour in the Marine Environment*. World Scientific, 2002.

Swan CC , Seo Y-K . Limit state analysis of earthen slopes using dual continuum/FEM approaches. *International Journal for Numerical and Analytical Methods in Geomechanics*. 1999;23(12):13591371.

R.G. Tart. Why the Oso Landslide Caused So Much Death and Destruction. In *Geotechnical and Structural Engineering Congress 2016*, volume 1, pages 15451554, Reston, VA, 2016. American Society of Civil Engineers.

Tatsuoka F . Discussion: The strength and dilatancy of sands. *Gotechnique*. 1987;37(2):219226.

Taylor DW . *Fundamentals of Soil Mechanics*. New York: Wiley; 1948.

Teh CI , Houlsby GT . Analytical study of the cone penetration test in clay. *Gotechnique*. 1991;41(1):1734.

F.S. Tehrani and V. Galavi. Comparison of Cavity Expansion and Material Point Method for Simulation of Cone Penetration in Sand. In *4th International Symposium on Cone Penetration Testing (CPT18)*, pages 611615, Delft, The Netherlands, 2018.

Terzaghi K . *Theorie der Setzung von Tonschichten : eine Einfuhrung in die analytische Tonmechanik* (in German). Leipzig: Franz Deuticke; 1936.

Terzaghi K . *Theoretical Soil Mechanics*. 1st ed. New York: John Wiley & Sons Inc; 1943.

Thilakasiri HS , Gunaratne M , Mullins G , Stinnette P , Jory B . Investigation of impact stresses induced in laboratory dynamic compaction of soft soils. *International Journal for Numerical and analytical methods in geomechanics*. 1996;20(10):753767.

Thompson EL , Huppert H . Granular column collapses: further experimental results. *Journal of Fluid Mechanics*. 2007;575:177186.

Thury M , Bossart P . Mont Terri rock laboratory: a new international research project in a Mesozoic shale formation in Switzerland. *Engineering Geology*. 1999;52:347359.

410R. Tielen, M. Mller, and C. Vuik. Efficient multigrid based solvers for isogeometric analysis. In *6th European Conference on Computational Mechanics and 7th European Conference on Computational Fluid Dynamics (ECCM-ECFD) Conference 2018*, 2018.

Tielen R , Wobbes E , Mller M , Beuth L . A high order material point method. *Procedia Engineering*. 2017;175:265272.

Tika TE , Hutchinson JN . Ring shear tests on soil from the Vaiont landslide slip surface. *Gotechnique*. 1999;49(1):5974.

Tolooiyan A , Gavin K . Modelling the cone penetration test in sand using cavity expansion and arbitrary Lagrangian-Eulerian finite element methods. *Computers and Geotechnics*. 2011;38(4):482490.

Truesdell C , Toupin RA . *The Classical Field Theories of Mechanics*. New York: Springer; 1960.

Ukritchon B , Whittle AJ , Sloan SW . Undrained stability of braced excavations in clay. *Journal of Geotechnical and Geoenvironmental Engineering*. 2003;129(8):738755.

S. Uriel and R. Molinia. Kinematic aspects of Vaiont slide. In *National Academy of Sciences, editor, 3rd International Conference of the ISRM*, volume 2B, pages 865870, Denver, USA, 1977.

van den Eijnden AP , Hicks MA . Efficient subset simulation for evaluating the modes of improbable slope failure. *Computers and Geotechnics*. 2017;88:267280.

van der Grinten JGM , van Dongen MEH , van der Kogel H . Strain and pore pressure propagation in a water-saturated porous medium. *Journal of Applied Physics*. 1987;62(12):46824687.

J.M. van Esch, D. Stolle, and I. Jassim. Finite element method for coupled dynamic flow-deformation simulation. In *2nd International Symposium in Computational Geomechanics (COMGEO II)*, pages 415524, Cavtat-Dubrovnik, Croatia, 2011.

Vardanega PJ , Stuart K . Haigh. The undrained strength-liquidity index relationship. *Canadian Geotechnical Journal*. 2014;200(1):40.

Vardoulakis I . Shear band inclination and shear modulus of sand in biaxial tests. *International Journal of Numerical and Analytical Methods in Geomechanics*. 1980;4(2):113119.

Vardoulakis I . Catastrophic landslides due to frictional heating of the failure plane. *Mechanics of Cohesive-Frictional Materials*. 2000;5(6):443467.

411 Vardoulakis I . Dynamic thermo-poro-mechanical analysis of catastrophic landslides. *Gotechnique*. *Gotechnique*. 2002;52(3):157171.

J. Vaunat, E. Romero, C. Marchi, and C. Jommi. Modelling of the shear strength of unsaturated soils. In *JFT Juc, TMP de Campos, and FAM Marinho, editors, 3rd International Conference of Unsaturated Soils, UNSAT 2002*, pages 245251. Swets & Zeitinger, 2002.

Verdugo R , Ishihara K . The steady state of sandy soils. *Soils and Foundations*. 1996;36(2):8191.

P.A. Vermeer, Y. Yuan, L. Beuth, and P. Bonnier. Application of interface elements with the Material Point Method. In *18th International Conference Computer Methods Mechanics*, pages 477478, 2009.

A. Veronese. *Rilievi sperimentali sugli sbocchi liberi*. *LEnergia Elettrica*, pages 441638, 1948.

Verruijt A . *An Introduction to Soil Dynamics*. Dordrecht: Springer; 2010.

Veveakis E , Vardoulakis I , Di Toro G . Thermoporemechanics of creeping landslides: The 1963 slide, northern Italy. *Journal of Geophysical Research*. 2007;112(F3):21562202.

Voight B , Faust C . Frictional heat and strength loss in some rapid slides. *Gotechnique*. 1982;32(1):4354.

Von Neumann J , Richtmyer RD . A method for the numerical calculation of hydrodynamic shocks. *Journal of Applied Physics*. 1950;21:232237.

Von Wolffersdorff PA . A hypoplastic relation for granular materials with a predefined limit state surface. *Mechanics of Cohesive-Frictional Materials*. 1996;1(3):25171.

Wallstedt PC , Guilkey JE . An evaluation of explicit time integration schemes for use with the generalized interpolation material point method. *Journal of Computational Physics*. 2008;227(22):96289642.

Wang B , Hicks MA , Vardon PJ . Slope failure analysis using the random material point method. *Geotechnics Letters*. 2016;6(2):113118.

Wang L , Coombs WM , Augarde CE , Brown M , Knappett J , Brennan A , Richards D , Blake A . Modelling screwpile installation using the MPM. *Procedia Engineering*. 2017;175:124132.

Wartman J , Montgomery DR , Anderson S , Keaton JR , Benot J , de la Chappelle J , Gilbert R . The 22 March 2014 Oso landslide, Washington, USA. *Geomorphology*. 2016;253(March):275288.

412 Wegener D , Herle I . Prediction of permanent soil deformations due to cyclic shearing with a hypoplastic constitutive model. *Geotechnik*. 2014;37(2):113122.

Wieckowski Z . Enhancement of the Material Point Method for Fluid-Structure Interaction and Erosion. Deltares Ltd, Delft: Technical report; 2013.

E. Wobbles, M. Mller, V. Galavi, and C. Vuik. Taylor least squares reconstruction for material point methods. In 6th European Conference on Computational Mechanics and 7th European Conference on Computational Fluid Dynamics (ECCM-ECFD) Conference 2018, 2018.

Wolter A , Stead D , Clague JJ . A morphologic characterisation of the 1963 Vajont Slide, Italy, using long-range terrestrial photogrammetry. *Geomorphology*. 2014;206:147164.

Wu Y , Chiew Y-M . Three-dimensional scour at submarine pipelines. *Journal of Hydraulic Engineering*. 2012;138(9):788795.

A. Yerro. MPM modelling of landslides in brittle and unsaturated soils. PhD thesis, Universitat Politcnica de Catalunya, 2015.

Yerro A , Alonso E , Pinyol N . Modelling large deformation problems in unsaturated soils. *E-UNSAT*. 2016;2016:16.

Yerro A , Alonso EE , Pinyol NM . The material point method for unsaturated soils. *Gotechnique*. 2015;65(3):201217.

Yerro A , Alonso EE , Pinyol NM . Run-out of landslides in brittle soils. *Computers and Geotechnics*. 2016;80:427439.

Yerro A , Pinyol NM , Alonso EE . Internal progressive failure in deep-seated landslides. *Rock Mechanics Rock Eng*. 2016;49(6):23172332.

A. Yerro, K. Soga, and J.D. Bray. Runout evaluation of the Oso landslide with the Material Point Method. *Canadian Geotechnical Journal*, accepted, 2018.

Yi JT , Goh SH , Lee FH , Randolph MF . A numerical study of cone penetration in fine-grained soils allowing for consolidation effects. *Gotechnique*. 2012;62(8):707.

A.R.II York, D.L. Sulsky, and H.L. Schreyer. The material point method for simulation of thin membranes. *International Journal of Numerical Methods in Engineering*, 44(10):14291456, 1999.

F. Zabala. Modelacin de problemas geotecnicos hidromecanicos utilizando el metodo del punto material. PhD thesis, Polytechnic University of Catalonia, 2010.

413 Zabala F , Alonso EE . Progressive failure of Aznalcollar dam using the material point method. *Gotechnique*. 2011;61(9):795808.

Zhang DZ , Ma X , Giguere PT . Material point method enhanced by modified gradient of shape function. *Journal of Computational Physics*. 2011;230(16):63796398.

X. Zhang, Z. Chen, and Y. Liu. *The Material Point Method: A Continuum-Based Particle Method*. Springer, 2017.

X. Zhao and D. Liang. MPM modelling of seepage flow through embankments. In Twenty-Sixth (2016) International Ocean and Polar Engineering Conference, pages 11611165, Rodos, Greece, 2016.

Zhao X , Liang D , Martinelli M . MPM simulations of dam-break floods. *Journal of Hydrodynamics, Ser. B*. 2017;29(3):397404.

Zheng H , Tham LG , Liu D . On two definitions of the factor of safety commonly used in the finite element slope stability analysis. *Computers and Geotechnics*. 2006;33(3):188195.

Zhu D , Griffiths DV , Huang J , Fenton GA . Probabilistic stability analyses of undrained slopes with linearly increasing mean strength. *Gotechnique*. 2017;67(8):733746.

Zienkiewicz OC , Chan AHC , Pastor M , Schrefler BA , Shiomi T . *Computational Geomechanics with Special Reference to Earthquake Engineering*. Chichester, UK: Wiley; 1999.

Zienkiewicz OC , Shiomi T . Dynamic behaviour of saturated porous media; the generalized Biot formulation and its numerical solution. *International Journal for Numerical and Analytical Methods in Geomechanics*. 1984;8(1):7896.

O.C. Zienkiewicz, Y.M. Xie, B.A. Schrefler, A. Ledesma, and N. Bicanic. *Static and Dynamic Behaviour of Soils: A Rational Approach to Quantitative Solutions. II. Semi-Saturated Problems*. Royal Society A: Mathematical, Physical and Engineering Sciences, 429(1877):311321, 1990.

B. Zuada Coelho, A. Rohe, A. Aboufirass, J. Nuttall, and M. Bolognin. Assessment of dike safety within the framework of large deformation analysis with the material point method. In A.S. Cardoso, J.L. Borges, P.A. Costa, A.T. Gomes, J.C. Marques, and C.S. Vieira, editors, *Numerical Methods in Geotechnical Engineering IX*, pages 657663. CRC Press, 2018.

B. Zuada Coelho, A. Rohe, and K. Soga. Poroelastic solid flow with double point material point method. *Journal of Hydrodynamics*, 29(3):423430, 2017.